Seminar on Algebraic Geometry 3

Exposé V: Construction of Quotient Schemes

English Translation and Native TeX Reconstruction — Loop 2

English reconstruction in progress

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Reconstruction note

The corrected Polo–Gille Exposé-V PDF controls the French text, numbering, formulas, notes, and diagram appearance. Because no editable Polo–Gille source was publicly available when this reconstruction began, Loop 1 reconstructed the formulas in LaTeX and used source-PDF PNG images as temporary diagram witnesses. This Loop 2 reader replaces those temporary images with native LaTeX diagrams checked against the corrected PDF while preserving the PNG witnesses outside the rendered reader.

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Exposé V
Construction of Quotient Schemes

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The purpose of this exposé is to prove the theorems stated in TDTE III.[1] If X and T are two objects of a category $\mathscr{C}$, we write $X(T)$ instead of $\mathrm{Hom}_{\mathscr{C}}(T, X)$. Similarly, if $\varphi : Y \to X$ is an arrow (respectively, if T is an object) of $\mathscr{C}$, then $\varphi(T)$ denotes the map $g \mapsto \varphi \circ g$ from $Y(T)$ to $X(T)$:

$$\begin{tikzcd} T \arrow[d, "g"'] \arrow[dr, dashed, "\varphi\circ g"] & \\ Y \arrow[r, "\varphi"'] & X \end{tikzcd}$$

Figure 1: The map $\varphi(T) : Y(T) \to X(T)$.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Polo–Gille Exp5-13oct24.pdf, component-local p. 1; combined-reader p. 263; printed p. 251.

Likewise, $T(\varphi)$ denotes the map $g \mapsto g \circ \varphi$ from $T(X)$ to $T(Y)$:

$$\begin{tikzcd} Y \arrow[r, "\varphi"] \arrow[dr, dashed, "g\circ\varphi"'] & X \arrow[d, "g"] \\ & T \end{tikzcd}$$

Figure 2: The map $T(\varphi) : T(X) \to T(Y)$.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Polo–Gille Exp5-13oct24.pdf, component-local p. 1; combined-reader p. 263; printed p. 251.

Finally, if P is a scheme, we denote by $\underline{P}$ the underlying set of P.

As an exception, in the present exposé we do not follow the convention on quotient notation stated in IV 4.6.15 (*loc. cit.*, top of p. 227 of the original), because we wish to give here a construction of quotients that also applies to "pre-equivalence relations"[2] that are not equivalence relations.

[1] Editor's note: namely, Theorems 5.1, 5.3, 6.1, 6.2, and 7.2 of TDTE III. The first two (respectively, the next two) correspond to Theorem 4.1 (respectively, Theorems 7.1 and 8.1) of this exposé. Theorem 7.2 of TDTE III is proved in Exposé $\mathrm{VI_A}$, 3.2 and 3.3.

[2] Editor's note: that is, groupoids with base X; see the terminology at the end of Section 1. When $\mathscr{C}$ is the

1. $\mathscr{C}$-groupoids

a) Let $\mathscr{C}$ be a category in which products and fiber products exist. Recall first that a diagram

$$X_1 \underset{d_0}{\overset{d_1}{\rightrightarrows}} X_0 \xrightarrow{p} Y$$

Figure 3: A parallel pair followed by its proposed cokernel.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Polo–Gille Exp5-13oct24.pdf, component-local p. 2; combined-reader p. 264; printed p. 252.

in $\mathscr{C}$ is said to be *exact* if $pd_0 = pd_1$ and if, for every $T \in \mathscr{C}$, the map $T(p)$ is a bijection from $T(Y)$ onto the subset of $T(X_0)$ consisting of the arrows $f : X_0 \to T$ such that $fd_0 = fd_1$. One also says that (Y, p) is the *cokernel* of (d_0, d_1), and writes

$$(Y, p) = \mathrm{Coker}(d_0, d_1).$$

b) For example, let $\mathscr{C}$ be the category (**Esp**.**An**) of ringed spaces. In this case a cokernel (Y, p) always exists, and it may be described as follows. The underlying topological space of Y is obtained from X_0 by identifying the points $d_0(x)$ and $d_1(x)$, and by giving Y the quotient topology. The canonical map $\pi : X_0 \to Y$, together with d_0, d_1, then induces a pair of arrows of sheaves of rings on Y:

$$\pi_*(\mathcal{O}_0) \underset{\delta_1}{\overset{\delta_0}{\rightrightarrows}} \pi_*(d_{0*}\mathcal{O}_1) = \pi_*(d_{1*}\mathcal{O}_1)$$

Figure 4: The pair of induced arrows between direct-image sheaves.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Polo–Gille Exp5-13oct24.pdf, component-local p. 2; combined-reader p. 264; printed p. 252.

Here $\mathcal{O}_i$ is the structure sheaf of X_i. As the sheaf of rings on Y, one chooses the subsheaf of $\pi_*(\mathcal{O}_0)$ whose sections s satisfy $\delta_0(s) = \delta_1(s)$. The arrow p is defined in the evident way.

Consider[3] the following diagram in (**Esp**.**An**):

$$X_1 \underset{d_0}{\overset{d_1}{\rightrightarrows}} X_0$$

Figure 5: The pair $(d_0, d_1) : X_1 \rightrightarrows X_0$.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Polo–Gille Exp5-13oct24.pdf, component-local p. 2; combined-reader p. 264; printed p. 252.

Let (Y, p) be its cokernel. An open subset U of X_0 is said to be *saturated* if

$$d_0^{-1}(U) = d_1^{-1}(U),$$

which is equivalent to saying $U = p^{-1}(p(U))$. In this case, since Y has the quotient topology, $p(U)$ is an open subset of Y.

category of schemes, the quotient $p : X \to Y$ of a groupoid X_* with base X exists under certain hypotheses (see 4.1, 6.1, and 7.1). If, in addition, X_* is an equivalence relation, then under the same hypotheses p is faithfully flat and quasi-compact, hence a universal epimorphism; see *loc. cit.*.

[3]Editor's note: Lemmas 1.1 and 1.2 have been added; they are used several times in Sections 5–9.

Lemma 1.1. *Let U be a saturated open subset of X, and set $V = p(U)$. Denote by $U_1 = d_0^{-1}(U) = d_1^{-1}(U)$ the corresponding open subset of X_1, and by $\widetilde{d}_0, \widetilde{d}_1$, and $\widetilde{p}$ the restrictions of d_0, d_1 to U_1, and of p to U, respectively. Then $(V, \widetilde{p})$ is a cokernel in $(\mathbf{Esp.An})$ of:*[4]

$$\begin{tikzcd} U_1 \arrow[r, shift left, "\widetilde{d}_1"] \arrow[r, shift right, "\widetilde{d}_0"'] & U \arrow[r, "\widetilde{p}"] & V \end{tikzcd}$$

Figure 6: The restricted pair and its cokernel in Lemma 1.1.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Polo–Gille Exp5-13oct24.pdf, component-local p. 2; combined-reader p. 264; printed p. 252.

The verification is easy.

Lemma 1.2.— *Let $d_0, d_1 : X_1 \rightrightarrows X_0$ be a diagram in $(\mathbf{Sch})$, and let (Y, p) be its cokernel in $(\mathbf{Esp.An})$.*

(i) If Y is a scheme and p is a morphism of schemes, then (Y, p) is a cokernel of (d_0, d_1) in $(\mathbf{Sch})$.

(ii) Suppose that every point of X_0 has a saturated open neighborhood U with the following property. Let $\widetilde{d}_0$ and $\widetilde{d}_1$ be the restrictions of d_0 and d_1 to $d_0^{-1}(U) = d_1^{-1}(U)$, and let (Q, q) be the cokernel of $(\widetilde{d}_0, \widetilde{d}_1)$ in $(\mathbf{Esp.An})$. If Q is a scheme and q is a morphism of schemes, then (Y, p) is a cokernel of (d_0, d_1) in $(\mathbf{Sch})$.

Part (i) is proved in §4(c); since the proof is short, let us repeat it here. Let $f : X_0 \to Z$ be a morphism of schemes such that $fd_0 = fd_1$. By hypothesis there is a unique morphism of ringed spaces $r : Y \to Z$ such that $f = rp$. It remains to show that, for every $y \in Y$, the homomorphism $\mathscr{O}_{r(y)} \to \mathscr{O}_y$ induced by r is local. This follows because p is surjective, so $y = p(x)$ for some x, and because the homomorphism $\mathscr{O}_{f(x)} \to \mathscr{O}_x$ induced by f is local.

Part (ii) follows from part (i) and the preceding lemma.

c) In this exposé we study the existence of $\mathrm{Coker}(d_0, d_1)$ when the double arrow (d_0, d_1) is placed in a richer context. More precisely, let

$$X_2 = X_1 \times_{d_1, d_0} X_1$$

be the fiber product of the diagram

$$\begin{tikzcd} & X_1 \arrow[d, "d_1"'] \\ X_1 \arrow[r, "d_0"'] & X_0 \end{tikzcd} \qquad (*)$$

[4] Editor's note: this is not the case in the category of schemes. For example, let $S = \mathrm{Spec}(\mathbf{C})$, let $X_0 = \mathbf{A}^2_S = \mathrm{Spec}(\mathbf{C}[x_1, x_2])$, let $d_1 : \mathbf{G}_{m,S} \times_S \mathbf{A}^2_S \to \mathbf{A}^2_S$ be the action of $\mathbf{G}_{m,S}$ on $\mathbf{A}^2_S$ by homotheties, let d_0 be the projection onto the second factor, and let $U = \mathbf{A}^2_S \setminus \{m\}$, where $m = (0, 0)$. Then the projective line $\mathbf{P}^1_S$ is the cokernel of $(\widetilde{d}_0, \widetilde{d}_1)$ in both $(\mathbf{Esp.An})$ and $(\mathbf{Sch})$, whereas the cokernel Y of (d_0, d_1) in $(\mathbf{Esp.An})$ is the union of $\mathbf{P}^1_S$ and the point $y_0 = \{p(m)\}$. The only open subset containing y_0 is Y, and $\Gamma(Y, \mathcal{O}_Y) = \mathbf{C}$. If $f : \mathbf{A}^2_S \to T$ is a morphism of S-schemes such that $fd_0 = fd_1$, and if $V = \mathrm{Spec}(A)$ is an affine open subset of T containing $t_0 = f(y_0)$, then $f^{-1}(V) = \mathbf{A}^2$, and the ring homomorphism $A \to \mathbf{C}[x_1, x_2]$ factors through $\mathbf{C}$. This shows that $S = \mathrm{Spec}(\mathbf{C})$ is the cokernel of (d_0, d_1) in the category $(\mathbf{Sch}/S)$.

and let $d_0', d_2' : X_2 \to X_1$ be the two canonical projections. Thus, by definition, the square

$$\begin{tikzcd} X_2 \arrow[r, "d_0'"] \arrow[d, "d_2'"'] & X_1 \arrow[d, "d_1"] \\ X_1 \arrow[r, "d_0"'] & X_0 \end{tikzcd} \tag{0}$$

is Cartesian.

In addition, suppose that a third arrow $d_1' : X_2 \to X_1$ is given. We say that

$$(d_0, d_1 : X_1 \rightrightarrows X_0, d_1')$$

is a *$\mathscr{C}$-groupoid* if, for every object T of $\mathscr{C}$, $X_1(T)$ is the set of arrows of a groupoid $X_*(T)$ whose set of objects is $X_0(T)$, whose source map is $d_1(T)$, whose target map is $d_0(T)$, and whose composition map is $d_1'(T)$. As usual, we identify

$$(X_1 \times_{d_1,d_0} X_1)(T) \simeq X_1(T) \times_{d_1(T),d_0(T)} X_1(T),$$

and recall that a groupoid is a category in which every arrow is invertible.[5]

If φ is an arrow of the groupoid $X_*(T)$, the map $f \mapsto \varphi \circ f$ is a bijection from the set of arrows f whose target equals the source of φ onto the set of arrows having the same target as φ. This is equivalently expressed by saying that the square

$$\begin{tikzcd} X_2 \arrow[r, "d_1'"] \arrow[d, "d_0'"'] & X_1 \arrow[d, "d_0"] \\ X_1 \arrow[r, "d_0"'] & X_0 \end{tikzcd} \tag{1}$$

is Cartesian.

Similarly, $g \mapsto g \circ \varphi$ is a bijection from the set of arrows g of $X_*(T)$ whose source equals the target of φ onto the set of arrows having the same source as φ. Equivalently, the square

$$\begin{tikzcd} X_2 \arrow[r, "d_1'"] \arrow[d, "d_2'"'] & X_1 \arrow[d, "d_1"] \\ X_1 \arrow[r, "d_1"'] & X_0 \end{tikzcd} \tag{2}$$

is Cartesian.

On the other hand, let $s : X_0 \to X_1$ be the unique arrow of $\mathscr{C}$ such that, for every T, the map $s(T) : X_0(T) \to X_1(T)$ sends each object of $X_*(T)$ to its identity arrow.[6] The arrow s satisfies

$$d_1 s = \mathrm{id}_{X_0}, \tag{3}$$
$$d_0 s = \mathrm{id}_{X_0}\,. \tag{3 bis}$$

[5]Editor's note: in this case $X_2(T)$ is the set of pairs (f_2, f_1) of composable arrows, that is, pairs such that $d_0(f_1) = d_1(f_2)$; the maps d_0', d_1', d_2' send (f_2, f_1), respectively, to $f_2, f_2 \circ f_1, f_1$.

[6]Editor's note: $T \mapsto s(T)$ defines an element of $\mathrm{Hom}(h_{X_0}, h_{X_1})$, and the latter is equal to $\mathrm{Hom}(X_0, X_1)$ by the Yoneda lemma.

Finally, associativity of the composition maps $d'_1(T)$ is equivalent to commutativity of the diagram

$$\begin{tikzcd} X_1 \times_{d_1,d_0} X_1 \times_{d_1,d_0} X_1 \arrow[r, "d'_1 \times X_1"] \arrow[d, "X_1 \times d'_1"'] & X_1 \times_{d_1,d_0} X_1 \arrow[d, "d'_1"'] \\ X_1 \times_{d_1,d_0} X_1 \arrow[r, "d'_1"'] & X_1 \end{tikzcd} \tag{4}$$

Conversely, conditions (1), (2), and (4), together with the existence of an arrow s satisfying (3), imply that $(X_1 \rightrightarrows X_0, d'_1)$ is a $\mathscr{C}$-groupoid. Condition (3) is harmless: it merely ensures that $d_1(T) : X_1(T) \to X_0(T)$ is surjective for every $T \in \mathscr{C}$. In the sequel we shall mainly use the Cartesian squares (0), (1), and (2), summarized in the diagram

$$\begin{tikzcd} X_2 \arrow[r, shift left, "d'_1"] \arrow[r, shift right, "d'_0"'] \arrow[d, "d'_2"'] & X_1 \arrow[r, "d_0"] \arrow[d, "d_1"'] & X_0 \\ X_1 \arrow[r, shift left, "d_1"] \arrow[r, shift right, "d_0"'] & X_0 & \end{tikzcd} \tag{0,1,2}$$

The two left-hand squares in this diagram (that is, squares (0) and (2)) are Cartesian; the first row is exact; and X_2 is identified with the fiber product $X_1 \times_{d_0,d_0} X_1$.

We use associativity only indirectly, for example to ensure the existence of an arrow s satisfying (3) and (3 bis), or to ensure the existence of an arrow

$$\sigma : X_1 \longrightarrow X_1, \qquad d_0\sigma = d_1, \quad d_1\sigma = d_0. \tag{†}$$

Here σ is chosen so that $\sigma(T) : X_1(T) \to X_1(T)$ sends every arrow of $X_*(T)$ to its inverse.[7]

By an abuse of language, we shall sometimes call a diagram

$$\begin{tikzcd} X_2 \arrow[r, shift left=2, "{d'_0,d'_1,d'_2}"] \arrow[r] \arrow[r, shift right=2] & X_1 \arrow[r, shift left, "{d_0,d_1}"] \arrow[r, shift right] & X_0 \end{tikzcd}$$

a $\mathscr{C}$-groupoid when (0), (1), and (2) are Cartesian, (4) is commutative, and there exists an s satisfying (3). Thus X_2 may be "a" fiber product of (*) without being "the" fiber product of (*).[8]

Terminology. In place of the $\mathscr{C}$-groupoid X_*, we shall also speak of the *groupoid X_* with base X_0*, or of the *pre-equivalence relation X_* in X_0*.

2. Examples of $\mathscr{C}$-groupoids

a) Let X be an object of $\mathscr{C}$, and let G be a $\mathscr{C}$-group acting on the left on X. We denote by $d_0 : G \times X \to X$ the arrow defining the action of G on X, by $d_1 : G \times X \to X$ the projection onto the second factor, and by $\mu : G \times G \to G$ the arrow defining

the $\mathscr{C}$-group structure on G, and finally by $\mathrm{pr}_{2,3}$ the projection of $G \times G \times X = G \times (G \times X)$ onto the second factor. Then

[7] Editor's note: the Yoneda lemma shows that σ is an involutive automorphism of X_1; this will be used, for example, in 3(e) and in Theorem 4.1.

[8] Editor's note: see Example 2(a) below.

$$G \times G \times X \overset{\mathrm{pr}_{2,3}}{\underset{G\times d_0}{\overset{\mu\times X}{\rightrightarrows}}} G \times X \overset{d_1}{\underset{d_0}{\rightrightarrows}} X$$

Figure 7: The groupoid attached to the action of the $\mathscr{C}$-group G on X.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Polo–Gille Exp5-13oct24.pdf, component-local p. 6; combined-reader p. 268; page header p. 256.

is a $\mathscr{C}$-groupoid.

b) Let $d_0, d_1 : X_1 \to X_0$ be an *equivalence pair*. In other words, if $d_0 \boxtimes d_1 : X_1 \to X_0 \times X_0$ is the arrow with components d_0 and d_1, we suppose that, for every object T of $\mathscr{C}$, the map $(d_0 \boxtimes d_1)(T)$ is a bijection from $X_1(T)$ onto the graph of an equivalence relation on $X_0(T)$. Consequently, $X_1(T)$ is identified with the set of pairs (x, y) of elements of $X_0(T)$ such that $x \sim y$. Likewise,

$$X_2(T) = \big(X_1 \times_{d_1,d_0} X_1\big)(T)$$

is identified with the set of triples (x, y, z) of elements of $X_0(T)$ such that $x \sim y$ and $y \sim z$. There is therefore a unique arrow $d_1' : X_2 \to X_1$ making squares (1) and (2) commute: $d_1'(T)$ must send $(x, y, z) \in X_2(T)$ to $(x, z) \in X_1(T)$. With this choice of d_1',

$$\big(d_0, d_1 : X_1 \rightrightarrows X_0, d_1'\big)$$

is a $\mathscr{C}$-groupoid.

Conversely, consider a $\mathscr{C}$-groupoid X_* for which $d_0 \boxtimes d_1 : X_1 \to X_0 \times X_0$ is a monomorphism. Then (d_0, d_1) is an equivalence pair, and X_* can be reconstructed from (d_0, d_1) as explained a few lines above.[9]

c) If $p : X \to Y$ is any arrow of $\mathscr{C}$, and if pr_1 and pr_2 are the two projections of $X \times_{p,p} X$ onto X, then

$$(\mathrm{pr}_1, \mathrm{pr}_2) : X \times_{p,p} X \rightrightarrows X$$

is an equivalence pair. The arrow p is called an *effective epimorphism* if the diagram

$$X \times_{p,p} X \overset{\mathrm{pr}_1}{\underset{\mathrm{pr}_2}{\rightrightarrows}} X \xrightarrow{p} Y$$

Figure 8: The defining diagram for an effective epimorphism.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Polo–Gille Exp5-13oct24.pdf, component-local p. 6; combined-reader p. 268; page header p. 256.

is exact, that is, if

$$(Y, p) = \mathrm{Coker}(\mathrm{pr}_1, \mathrm{pr}_2).$$

For example, let S be a Noetherian scheme and let $\mathscr{C}$ be the category of schemes finite over S. We show that an epimorphism in $\mathscr{C}$ need not be effective. Take

$$S = \mathrm{Spec}\, k[T^3, T^5], \qquad Y = S, \qquad X = \mathrm{Spec}\, k[T],$$

where k is a commutative field. If i is the inclusion of $B = k[T^3, T^5]$ into $A = k[T]$, take $p = \mathrm{Spec}\, i$. Then $X \times_{p,p} X$ is identified with $\mathrm{Spec}(A \otimes_B A)$, and $\mathrm{Coker}(\mathrm{pr}_1, \mathrm{pr}_2)$ with $\mathrm{Spec}\, B'$, where B' is the subring of A consisting of the elements a such that $a \otimes_B 1 = 1 \otimes_B a$. But

$$T^7 \otimes_B 1 = (T^2 T^5) \otimes_B 1 = T^2 \otimes_B T^5 = T^2 \otimes_B (T^3 T^2) = T^5 \otimes_B T^2 = 1 \otimes_B T^7.$$

[9] Editor's note: in particular, if G is a $\mathscr{C}$-group acting on the left on an object X of $\mathscr{C}$, and if X_* is the $\mathscr{C}$-groupoid defined in 2(a), then (d_0, d_1) is an equivalence pair if and only if G acts freely on X; cf. Exposé III, 3.2.1.

Thus T^7 belongs to B', does not belong to B, and $\operatorname{Spec} B'$ is different from $\operatorname{Spec} B$, which gives the counterexample.[10]

3. Some sorites concerning $\mathscr{C}$-groupoids

Here are, in no particular order, some remarks used below.

a) Let

$$X_2 \overset{d'_0,d'_1,d'_2}{\Rrightarrow} X_1 \overset{d_0,d_1}{\rightrightarrows} X_0$$

Figure 9: The $\mathscr{C}$-groupoid X_*.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Polo–Gille Exp5-13oct24.pdf, component-local p. 7; combined-reader p. 269; page header p. 257.

be a $\mathscr{C}$-groupoid, and let $f_0 : Y_0 \to X_0$ be an arrow of $\mathscr{C}$. We shall define a $\mathscr{C}$-groupoid with base Y_0,

$$Y_2 \overset{e'_0,e'_1,e'_2}{\Rrightarrow} Y_1 \overset{e_0,e_1}{\rightrightarrows} Y_0$$

Figure 10: The induced $\mathscr{C}$-groupoid Y_*.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Polo–Gille Exp5-13oct24.pdf, component-local p. 7; combined-reader p. 269; page header p. 257.

which will be said to be *induced by X_* and f_0*. We also say that Y_* is the *inverse image of X_* under the base change f_0*.

We choose Y_1 to be the fiber product of the diagram

$$\begin{tikzcd} Y_1 \arrow[r, dashed, "f_1"] \arrow[d, dashed] & X_1 \arrow[d, "d_0 \boxtimes d_1"] \\ Y_0 \times Y_0 \arrow[r, "f_0 \times f_0"'] & X_0 \times X_0 \end{tikzcd}$$

Figure 11: The fiber-product definition of Y_1.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Polo–Gille Exp5-13oct24.pdf, component-local p. 7; combined-reader p. 269; page header p. 257.

and take e_0 and e_1 to be the composites of the canonical arrow $Y_1 \to Y_0 \times Y_0$ with the first and second projections of $Y_0 \times Y_0$, respectively. The morphism $Y_1 \to Y_0 \times Y_0$ is then $e_0 \boxtimes e_1$, and

$$f_0 \circ e_i = d_i \circ f_1 \qquad (i = 0, 1),$$

where f_1 denotes the projection of Y_1 onto X_1.

Set $Y_2 = Y_1 \times_{e_0,e_1} Y_1$, cf. 1(c). One may say that the pair (e_0, e_1) is defined so that, for every $T \in \mathscr{C}$ and every pair (y, x) of elements of $Y_0(T)$, there is a certain bijective correspondence

$$\psi \longmapsto {}_y\psi_x$$

[10]Editor's note: the same argument applies to $B = k[T^3, T^4]$ and $T^5 \otimes_B 1$; more generally, it applies to $B = k[T^n, T^{n+r}]$ and the element $T^{n+2r} \otimes_B 1$, provided that n does not divide $2r$.

between the arrows ψ of $X_*(T)$ with source $f_0(x)$ and target $f_0(y)$, and the arrows ${}_y\psi_x$ of $Y_*(T)$ with source x and target y. Thus $e_1' : Y_2 \to Y_1$ is determined by defining, for every $T \in \mathscr{C}$, the composition of arrows in $Y_*(T)$ by the formula

$${}_z\psi_y \circ {}_y\varphi_x = {}_z(\psi \circ \varphi)_x.$$

It is clear that this definition makes each $Y_*(T)$ a groupoid.

b) Given the $\mathscr{C}$-groupoid X_* and the base change $f_0 : Y_0 \to X_0$, one can reconstruct the pair $(e_0, e_1) : Y_1 \rightrightarrows Y_0$ in another way.[11] Construct $Y_0 \times_{X_0} X_1$, pr_1, and pr_2 so that the square

$$\begin{tikzcd}
Y_0 \times_{X_0} X_1 \arrow[r, "\mathrm{pr}_2"] \arrow[d, "\mathrm{pr}_1"'] & X_1 \arrow[d, "d_0"'] \\
Y_0 \arrow[r, "f_0"'] & X_0
\end{tikzcd}$$

Figure 12: The Cartesian square defining $Y_0 \times_{X_0} X_1$.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Polo–Gille Exp5-13oct24.pdf, component-local p. 7; combined-reader p. 269; page header p. 257.

is Cartesian. A straightforward reduction to the set-theoretic case then shows that the following square is Cartesian:

$$\begin{tikzcd}
Y_1 \arrow[r, "e_0 \boxtimes f_1"] \arrow[d, "e_1"'] & Y_0 \times_{X_0} X_1 \arrow[d, "d_1 \circ \mathrm{pr}_2"] \\
Y_0 \arrow[r, "f_0"'] & X_0
\end{tikzcd}$$

Figure 13: The second Cartesian square in the reconstruction of (e_0, e_1).

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Polo–Gille Exp5-13oct24.pdf, component-local p. 8; combined-reader p. 270; page header p. 258.

Here f_1 denotes the canonical projection of

$$Y_1 = (Y_0 \times Y_0) \times_{(X_0 \times X_0)} X_1$$

onto X_1.

c) We give two examples of inverse images of a $\mathscr{C}$-groupoid. Take $Y_0 = X_1$ and $f_0 = d_0$. For every object T of $\mathscr{C}$, the set $Y_1(T)$ is then identified with the set of diagrams of the form

$$\begin{tikzcd}
b \arrow[r, "\varphi"] & d \\
a \arrow[u, "f"] & c \arrow[u, "g"']
\end{tikzcd}$$

Figure 14: An arrow of the first inverse-image groupoid.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Polo–Gille Exp5-13oct24.pdf, component-local p. 8; combined-reader p. 270; page header p. 258.

[11] Editor's note: this second viewpoint is used in 3(f) and in the proof of 6.1.

in $X_*(T)$. The source of such a diagram is the arrow f, and its target is the arrow g. These diagrams compose in the evident way.

Now set $Y_0' = X_1$ and $f_0' = d_1$ (we add the primes[12] to avoid any confusion with the preceding example). In this case, for every $T \in \mathscr{C}$, the set $Y_1'(T)$ is identified with the set of diagrams of the form

$$\begin{tikzcd} b & d \\ a \arrow[u, "f"] \arrow[r, "\psi"'] & c \arrow[u, "g"'] \end{tikzcd}$$

Figure 15: An arrow of the second inverse-image groupoid.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Polo–Gille Exp5-13oct24.pdf, component-local p. 8; combined-reader p. 270; page header p. 258.

in the groupoid $X_*(T)$. The source of such a diagram is f, its target is g, and these diagrams again compose in the evident way.

It is now clear that the identity map of $Y_0(T)$, together with the map

$$\begin{tikzcd} b \arrow[r, "\varphi"] & d \\ a \arrow[u, "f"] & c \arrow[u, "g"'] \end{tikzcd} \quad \mapsto \quad \begin{tikzcd} b & d \\ a \arrow[u, "f"] \arrow[r, "g^{-1}\circ\varphi\circ f"'] & c \arrow[u, "g"'] \end{tikzcd}$$

Figure 16: The map from $Y_1(T)$ to $Y_1'(T)$.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Polo–Gille Exp5-13oct24.pdf, component-local p. 8; combined-reader p. 270; page header p. 258.

defines an isomorphism from the groupoid $Y_*(T)$ onto $Y_*'(T)$. Moreover, this isomorphism depends functorially on T, so the $\mathscr{C}$-groupoids Y_* and Y_*' are isomorphic.[13]

d)

Proposition 3.1.— *Keep the notation of 3(a), and suppose that f_0 is an effective and universal epimorphism. Then* $\mathrm{Coker}(d_0, d_1)$ *exists if and only if* $\mathrm{Coker}(e_0, e_1)$ *exists.*[14] *Moreover, in this case f_0 induces an isomorphism*

$$\mathrm{Coker}(d_0, d_1) \xrightarrow{\sim} \mathrm{Coker}(e_0, e_1).$$

Recall first that an epimorphism $f_0 : Y_0 \to X_0$ is called *universal* if, in every Cartesian square

$$\begin{tikzcd} Y' \arrow[r] \arrow[d, "f'"'] & Y_0 \arrow[d, "f_0"] \\ X' \arrow[r] & X_0 \end{tikzcd}$$

Figure 17: A base change of the universal epimorphism f_0.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Polo–Gille Exp5-13oct24.pdf, component-local p. 9; combined-reader p. 271; page header p. 259.

[12]Editor's note: "accents" in the original.

[13]Editor's note: this plays a crucial role in the proof of Lemma 6.1.

[14]Editor's note: the original has been modified to display explicitly the isomorphism below.

the arrow f' is an epimorphism. Now let $C(d_0, d_1)$ denote the covariant functor from $\mathscr{C}$ to sets which assigns to every $T \in \mathscr{C}$ the kernel of the pair

$$T(d_0), T(d_1) : T(X_0) \rightrightarrows T(X_1),$$

and define $C(e_0, e_1)$ in the same way. For every $T \in \mathscr{C}$, there is therefore a commutative diagram

$$\begin{tikzcd}
C(d_0,d_1)(T) \arrow[r] \arrow[d, "T(f)"'] & T(X_0) \arrow[r, shift left, "T(d_1)"] \arrow[r, shift right, "T(d_0)"'] \arrow[d, "T(f_0)"'] & T(X_1) \arrow[d, "T(f_1)"'] \\
C(e_0,e_1)(T) \arrow[r] & T(Y_0) \arrow[r, shift left, "T(e_1)"] \arrow[r, shift right, "T(e_0)"'] & T(Y_1)
\end{tikzcd}$$

Figure 18: The comparison of the functors $C(d_0, d_1)$ and $C(e_0, e_1)$.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Polo–Gille Exp5-13oct24.pdf, component-local p. 9; combined-reader p. 271; page header p. 259.

Here $T(f)$ is the injection induced by the injection $T(f_0)$. If we show that $T(f)$ is surjective for every T, we obtain a functorial isomorphism

$$f : C(d_0, d_1) \xrightarrow{\sim} C(e_0, e_1).$$

Thus one of these functors is representable if and only if the other is, which proves the proposition.

To prove that $T(f)$ is surjective, consider the diagram

$$\begin{tikzcd}
 & Y_1 \arrow[r, "f_1"] \arrow[d, shift right, "e_0"'] \arrow[d, shift left, "e_1"] & X_1 \arrow[d, shift right, "d_0"'] \arrow[d, shift left, "d_1"] \\
Y_0 \times_{X_0} Y_0 \arrow[ur, "\Delta"] \arrow[r, shift left, "\mathrm{pr}_2"] \arrow[r, shift right, "\mathrm{pr}_1"'] & Y_0 \arrow[r, "f_0"'] & X_0
\end{tikzcd}$$

Figure 19: The diagram used to prove the surjectivity of $T(f)$.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Polo–Gille Exp5-13oct24.pdf, component-local p. 9; combined-reader p. 271; page header p. 259.

Here Δ is the section of $Y_1 \to Y_0 \times Y_0$ defined by the morphism

$$s \circ f_0 \circ \mathrm{pr}_1 : Y_0 \times Y_0 \longrightarrow X_1,$$

where $s : X_0 \to X_1$ satisfies (3) and (3 bis) of Section 1.

If $g : Y_0 \to T$ satisfies $g \circ e_0 = g \circ e_1$, then

$$g \circ e_0 \circ \Delta = g \circ e_1 \circ \Delta,$$

so $g \circ \mathrm{pr}_1 = g \circ \mathrm{pr}_2$. Since f_0 is an effective epimorphism, g factors as $h \circ f_0$ for an arrow $h : X_0 \to T$; in other words, $g = T(f_0)(h)$. It remains to prove that $h \in C(d_0, d_1)(T)$, or equivalently that $hd_0 = hd_1$. But

$$hd_0 f_1 = hf_0 e_0 = ge_0 = ge_1 = hf_0 e_1 = hd_1 f_1,$$

and the desired equality follows because f_1 is an epimorphism (since f_0 is a universal epimorphism).

e) Now let S be a scheme and take $\mathscr{C} = (\mathbf{Sch}/S)$. The data of a $\mathscr{C}$-groupoid

$$X_2 \overset{d_2'}{\underset{d_0'}{\overset{d_1'}{\rightrightarrows\!\!\!\to}}} X_1 \underset{d_0}{\overset{d_1}{\rightrightarrows}} X_0$$

Figure 20: A $(\mathbf{Sch}/S)$-groupoid.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Polo–Gille Exp5-13oct24.pdf, component-local p. 10; combined-reader p. 272; page header p. 260.

defines an equivalence relation on the set $\underline{X_0}$ underlying the scheme X_0: for $x, y \in \underline{X_0}$, write $x \sim y$ if there exists $z \in \underline{X_1}$ such that $x = d_1 z$ and $y = d_0 z$. Reflexivity and symmetry are clear.[15] Let us prove transitivity. If $x \sim y$ and $y \sim z$, there exist $u, v \in \underline{X_1}$ such that

$$x = d_1 u, \qquad y = d_0 u, \qquad y = d_1 v, \qquad z = d_0 v.$$

It follows that (v, u) belongs to the set-theoretic fiber product $\underline{X_1} \times_{\underline{d_1}, \underline{d_0}} \underline{X_1}$. Since the canonical map

$$\underline{X_1 \times_{d_1, d_0} X_1} \longrightarrow \underline{X_1} \times_{\underline{d_1}, \underline{d_0}} \underline{X_1}$$

Figure 21: The canonical map from the underlying set of the fiber product to the fiber product of the underlying sets.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Polo–Gille Exp5-13oct24.pdf, component-local p. 10; combined-reader p. 272; page header p. 260.

is surjective, (v, u) is the image of some $w \in \underline{X_2}$. Hence

$$x = d_1 d_1' w, \qquad z = d_0 d_1' w,$$

and therefore $x \sim z$.

f) Keep the notation of 3(a) and 3(b), still taking $\mathscr{C} = (\mathbf{Sch}/S)$. If x, y are points of Y_0, then

$$x \sim y \quad \Longleftrightarrow \quad f_0(x) \sim f_0(y).$$

Thus the inverse image of the equivalence relation defined by a groupoid is the equivalence relation defined by the inverse image of that groupoid.

Indeed, suppose that $x \sim y$. There is a $z \in \underline{Y_1}$ such that $x = e_1(z)$ and $y = e_0(z)$. Since $f_0 \circ e_i = d_i \circ f_1$ for $i = 0, 1$, it follows that

$$f_0(x) = d_1 f_1(z), \qquad f_0(y) = d_0 f_1(z),$$

and hence $f_0(x) \sim f_0(y)$.

Conversely, suppose that $f_0(x) \sim f_0(y)$, and let $z \in \underline{X_1}$ satisfy

$$f_0(y) = d_1(z), \qquad f_0(x) = d_0(z).$$

[15] Editor's note: reflexivity follows from the existence of $s : X_0 \to X_1$, which is a section of both d_0 and d_1; symmetry follows from the existence of the involution σ of X_1 which "exchanges d_0 and d_1", that is, which satisfies $d_0\sigma = d_1$ and $d_1\sigma = d_0$; cf. Section 1, (3), (3 bis), and (†).

Using the construction and notation of 3(b), there is a point $t \in \underline{Y_0 \times_{X_0} X_1}$ such that $\mathrm{pr}_1(t) = x$ and $\mathrm{pr}_2(t) = z$. Since $f_0(y) = d_1 \,\mathrm{pr}_2(t)$, there is likewise an $s \in \underline{Y_1}$ such that

$$y = e_1(s), \qquad (e_0 \boxtimes f_1)(s) = t.$$

It follows that

$$e_0(s) = \mathrm{pr}_1(e_0 \boxtimes f_1)(s) = \mathrm{pr}_1(t) = x,$$

so $x \sim y$.

4. Passage to the quotient by a finite flat groupoid (proof of a special case)

Theorem 4.1.— *Consider a* $(\mathbf{Sch}/S)$*-groupoid*

$$\begin{tikzcd} X_2 \arrow[r, shift left=2, "d'_2"] \arrow[r, "d'_1" description] \arrow[r, shift right=2, "d'_0"'] & X_1 \arrow[r, shift left, "d_1"] \arrow[r, shift right, "d_0"'] & X_0 \end{tikzcd}$$

and suppose that the following conditions hold:[16]

a) d_1 *is finite locally free;*

b) for every $x \in X_0$*, the set* $d_0 d_1^{-1}(x)$ *is contained in an affine open subset of* X_0*.*[17]

Then:

(i) *there exists a cokernel* (Y, p) *of* (d_0, d_1) *in* $(\mathbf{Sch}/S)$*. Moreover, such a pair* (Y, p) *is a cokernel of* (d_0, d_1) *in the category of all ringed spaces;*

(ii) p *is integral and open, and* Y *is affine when* X_0 *is affine;*[18]

(iii) *the morphism*

$$X_1 \longrightarrow X_0 \times_Y X_0$$

with components d_0 *and* d_1 *is surjective;*

(iv) *if* (d_0, d_1) *is an equivalence pair, then* $X_1 \to X_0 \times_Y X_0$ *is an isomorphism*[19] *and* $p : X_0 \to Y$ *is finite locally free.*[20] *Moreover,* (Y, p) *is a cokernel of* (d_0, d_1) *in the category of sheaves for the fppf topology and, for every base change* $Y' \to Y$*, the scheme* Y' *is the cokernel of the groupoid* $X_* \times_Y Y'$ *obtained from* X_* *by the base change* $X_0 \times_Y Y' \to X_0$*.*

[16] Editor's note: since $d_0 = d_1\sigma$, where σ is an involutive automorphism of X_1, these two conditions are symmetric in d_1 and d_0. Moreover, $d_0 d_1^{-1}(x) = d_1 d_0^{-1}(x)$.

[17] Editor's note: condition b) cannot be omitted. H. Hironaka gave an example of an action of the finite group $G = \mathbb{Z}/2\mathbb{Z}$ on a proper complex variety X_0 for which the quotient X_0/G is an algebraic space that is not a scheme; see [Hi62], and also [Mum65], Chapter 4, §3.

[18] Editor's note: the assertion that p is open has been added by repeating the analogous argument given in 6.1.

[19] Editor's note: in this case $X_1 \to X_0 \times_S X_0$ is therefore an immersion (EGA I, 5.3.10); see also Exposé $\mathrm{VI_B}$, 9.2.1. For existence of the quotient—in the category of schemes or in that of algebraic spaces—under the weaker hypothesis that d_0 and d_1 are finite but not necessarily flat, see [An73], §1.1, [Fe03], [Ko08], and the references there.

[20] Editor's note: the consequences that follow have been made explicit; see [Ray67a], Theorem 1(iii), and the proof given below at the end of Section 5.

In particular, for every base change $S' \to S$, the scheme $Y' = Y \times_S S'$ is the cokernel of the S'-groupoid $X'_* = X_* \times_S S'$. Thus, in this case, formation of the quotient commutes with base change.

It follows immediately from (i) that the underlying topological space of Y is the quotient of the underlying topological space of X_0 by the equivalence relation defined by the $(\mathbf{Sch}/S)$-groupoid X_*.

We shall first prove the theorem when X_0 is affine and d_1 is locally free of constant rank n. We shall then see how to reduce the general situation to this special case.

In the case now under consideration, X_0, X_1, X_2 are affine. We may therefore suppose that

$$X_i = \operatorname{Spec} A_i, \qquad d_j = \operatorname{Spec} \delta_j, \qquad d'_k = \operatorname{Spec} \delta'_k,$$

where the A_i are commutative rings and the δ_j, δ'_k are ring homomorphisms. The diagram (0,1,2) may then be replaced by the following dual diagram:

$$(0,1,2)^* \qquad \begin{tikzcd} A_2 & A_1 \arrow[l, shift right, "\delta'_1"'] \arrow[l, shift left, "\delta'_0"] & A_0 \arrow[l, "\delta_0"'] \\ A_1 \arrow[u, "\delta'_2"] & A_0 \arrow[l, shift right, "\delta_1"'] \arrow[l, shift left, "\delta_0"] \arrow[u, "\delta_1"'] & \end{tikzcd}$$

The two left-hand squares in this diagram are cocartesian.

Let B be the subring of A_0 consisting of the elements $a \in A_0$ for which $\delta_0(a) = \delta_1(a)$.

a) We first show that A_0 is integral over B. For $a \in A_0$, let

$$P_{\delta_1}(T, \delta_0(a)) = T^n - \sigma_1 T^{n-1} + \cdots + (-1)^n \sigma_n \tag{1}$$

be the characteristic polynomial of $\delta_0(a)$, where A_1 is regarded as an A_0-algebra through δ_1; see Bourbaki, *Algèbre*, VIII, §12, and *Algèbre commutative*, II, §5, Exercise 9. Since the two left-hand squares of the diagram $(0,1,2)^*$ are cocartesian, one has

$$\delta_0(P_{\delta_1}(T, \delta_0(a))) = P_{\delta'_2}(T, \delta'_0\delta_0(a)), \tag{2}$$

$$\delta_1(P_{\delta_1}(T, \delta_0(a))) = P_{\delta'_2}(T, \delta'_1\delta_0(a)). \tag{3}$$

Because $\delta'_0\delta_0 = \delta'_1\delta_0$, it follows that

$$\delta_0(P_{\delta_1}(T, \delta_0(a))) = \delta_1(P_{\delta_1}(T, \delta_0(a)))\,;$$

equivalently,

$$\delta_0(\sigma_i) = \delta_1(\sigma_i) \qquad \text{for every } i.$$

On the other hand, the Cayley–Hamilton theorem gives

$$\delta_0(a)^n - \delta_1(\sigma_1)\delta_0(a)^{n-1} + \cdots + (-1)^n \delta_1(\sigma_n) = 0.$$

Since $\delta_1(\sigma_i) = \delta_0(\sigma_i)$, one also has

$$\delta_0(a)^n - \delta_0(\sigma_1)\delta_0(a)^{n-1} + \cdots + (-1)^n \delta_0(\sigma_n) = 0.$$

Hence

$$a^n - \sigma_1 a^{n-1} + \cdots + (-1)^n \sigma_n = 0,$$

because there is a homomorphism $\tau : A_1 \to A_0$ such that $\tau\delta_0 = \mathrm{id}_{A_0}$, so δ_0 is injective. It follows that A_0 is integral over B.

b) Consider two prime ideals x and y of A_0. We shall show that the equality

$$x \cap B = y \cap B$$

implies the existence of a prime ideal z of A_1 such that

$$x = d_0(z), \qquad y = d_1(z).$$

Indeed, if this assertion were false, x would be distinct from $\delta_0^{-1}(t)$ for every prime ideal t of A_1 such that $\delta_1^{-1}(t) = y$. For every such t, one would have

$$\delta_0^{-1}(t) \cap B = \delta_1^{-1}(t) \cap B = y \cap B = x \cap B.$$

It would follow from Cohen–Seidenberg (see Bourbaki, *Algèbre commutative*, V, §2, Corollary 1 to Proposition 1) that x is contained in none of the ideals $\delta_0^{-1}(t)$.[21] There are at most n prime ideals t of A_1 satisfying $\delta_1^{-1}(t) = y$ (see *loc. cit.*, Proposition 3). Hence, by the Prime Avoidance Lemma (*loc. cit.*, II, §1, Proposition 3), there would be an element $a \in x$ belonging to none of the ideals $\delta_0^{-1}(t)$.

Consequently, $\delta_0(a)$ would belong to none of these prime ideals t, and the lemma below would imply that the norm $N_{\delta_1}(\delta_0(a))$ does not belong to y. Here the norm is computed by regarding A_1 as an A_0-algebra through δ_1; with the notation of part a),

$$N_{\delta_1}(\delta_0(a)) = \sigma_n.$$

On the other hand,

$$(-1)^{n-1}\sigma_n = a^n + \sum_{i=1}^{n-1}(-1)^i\sigma_i a^{n-i},$$

so this norm belongs to $B \cap x = B \cap y$, a contradiction.

Lemma 4.1.1.— *Let $A \to A'$ be a homomorphism of commutative rings which makes A' a projective A-module of rank n. Let $p \in \mathrm{Spec}(A)$, let $q_1, \dots, q_r$ be the points of $\mathrm{Spec}(A')$ lying above p, and let $a \in A'$. Then $a \in q_1 \cup \cdots \cup q_r$ if and only if its norm $N(a)$ belongs to p.*

Indeed, after replacing A and A' by the localizations A_p and A'_p, we reduce to the case in which (A, p) is local and A' is semilocal, with

$$\mathrm{Spec}(A') = \{q_1, \dots, q_r\}.$$

In this case A' is a free A-module of rank n (see Bourbaki, *Algèbre commutative*, II, §3.2, Corollary 2 to Proposition 5), and $N(a)$ is the determinant of the endomorphism

$$\ell_a : A' \longrightarrow A', \qquad a' \longmapsto aa'.$$

Thus one has the equivalences

$$N(a) \notin p \iff N(a) \text{ is invertible} \iff \ell_a \text{ is invertible} \iff a \notin q_1 \cup \cdots \cup q_r.$$

[21] Editor's note: the original has been expanded in what follows; in particular, Lemma 4.1.1, taken from [DG70], III, §2.4, Lemma 4.3, has been added.

c) Proof of (i). Set $Y = \operatorname{Spec} B$ and $p = \operatorname{Spec} i$, where $i : B \hookrightarrow A_0$ is the inclusion. By part a), the morphism $p : X_0 \to Y$ is surjective. We first show that (Y, p) is a cokernel of (d_0, d_1) in the category of all ringed spaces. Indeed, part b) shows that the set underlying $\operatorname{Spec} B$ is obtained from the set underlying X_0 by identifying the points x and y for which there exists $z \in X_1$ such that

$$d_1 z = y, \qquad d_0 z = x.$$

Moreover, since i is integral, $p = \operatorname{Spec} i$ is closed, and therefore Y carries the quotient topology induced by X_0.

It follows that p is open. Indeed, let U' be any open subset of X_0. Since d_1 is surjective and finite locally free, it is faithfully flat and of finite presentation, hence open. Thus

$$U = d_1\big(d_0^{-1}(U')\big)\,,$$

the saturation of U' for the equivalence relation defined by X_*, is open. Since Y has the quotient topology, $p(U') = p(U)$ is open.

Finally, the choice of B, together with the fact that p, d_0, and d_1 are affine, shows that the canonical sequence of sheaves of rings

$$\mathcal{O}_Y \longrightarrow p_*\mathcal{O}_{X_0} \underset{p_*(\delta_0)}{\overset{p_*(\delta_1)}{\rightrightarrows}} p_*d_{0*}\mathcal{O}_{X_1} = p_*d_{1*}\mathcal{O}_{X_1}$$

is exact.

It remains to show that (Y, p) is also the cokernel of (d_0, d_1) in the category of schemes—more generally, in the category of locally ringed spaces. Let $q : X_0 \to Z$ be a morphism of schemes such that $qd_0 = qd_1$. By what precedes, there is a unique morphism of ringed spaces $r : Y \to Z$ such that $q = rp$. We must show that, for every $y \in Y$, the homomorphism

$$\mathcal{O}_{Z,r(y)} \longrightarrow \mathcal{O}_{Y,y}$$

induced by r is local. Since p is surjective, $y = p(x)$ for some $x \in X_0$, and the assertion follows from the fact that the homomorphism

$$\mathcal{O}_{Z,q(x)} \longrightarrow \mathcal{O}_{X_0,x}$$

induced by q is local.

d) Proof of (ii). This follows from parts a) and c).

e) Proof of (iii). Recall that $\underline{P}$ denotes the set underlying a scheme P, and that $\underline{d} : \underline{P} \to \underline{Q}$ denotes the map induced by a morphism $d : P \to Q$.

Lemma 4.1.2.—[22] *Let $(A, \mathfrak{m})$ be a local ring, let k be its residue field, and let K/k be a field extension. There exists a local, flat A-algebra B such that $B/\mathfrak{m}B$ is k-isomorphic to K. If K/k is finite, B may moreover be chosen finite and free over A.*

This is proved in EGA 0_{III}, 10.3.1, where it is also shown that B may be chosen Noetherian when A is Noetherian. For the reader's convenience, we recall the proof.

Put $A' = A[T]$, where T is an indeterminate. If $K = k(T)$, let $\mathfrak{p} = \mathfrak{m}A'$ and $B = A'_{\mathfrak{p}}$. Then

$$B/\mathfrak{m}B \simeq k[T]_{(0)} = k(T).$$

The ring B is flat over A', which is a free A-module, so B is flat over A.

[22]Editor's note: this lemma, which is used repeatedly in this Exposé and in later Exposés ($\mathrm{VI_A}$, $\mathrm{VI_B}$), has been inserted here. It appeared as Lemma $\mathrm{VI_B}$, 4.5.1, in the original 1965 edition of SGA 3.

Suppose next that $K = k(t) = k[t]$, where t is algebraic over k. Set $B = A'/(F)$, where $F \in A'$ is a monic polynomial whose image in $k[T]$ is the minimal polynomial f of t over k. Then B is a free A-module of finite rank

$$\deg(F) = \deg(f).$$

In particular, B is integral over A, and hence every maximal ideal of B contains $\mathfrak{m}$. Since

$$B/\mathfrak{m}B \simeq k[T]/(f) \simeq K,$$

B is local, with maximal ideal $\mathfrak{m}B$. This already proves that B can be chosen finite and free over A when $[K : k] < \infty$.

In the general case, choose a family $(t_i)_{i\in I}$ of generators of K/k, and well-order I; that is, give I a total order $\leq$ in which every nonempty subset has a least element. For $i \in I$, let k_i (respectively $k_{<i}$) be the subfield of K generated by the t_j with $j \leq i$ (respectively $j < i$). By adding one element if necessary, we may suppose that I has a greatest element ξ, so that $K = k_\xi$.

Let $J \subseteq I$ be the set of indices i for which there exists a direct system $(A_j)_{j\leq i}$ of local, flat A-algebras such that

$$A_j/\mathfrak{m}A_j \simeq k_j$$

and A_j is flat over A_ℓ whenever $\ell < j$. Suppose that $I \setminus J$ is nonempty, let i be its least element, and put

$$A' = \varinjlim_{j<i} A_j.$$

Since tensor products commute with direct limits, A' is flat over A and over every A_j, $j < i$, and

$$A'/\mathfrak{m}A' \simeq A' \otimes_A (A/\mathfrak{m}) \simeq k_{<i}.$$

Moreover, A' is local, with maximal ideal $\mathfrak{m}A'$. Indeed, if $x = f_j(x_j)$ is not invertible, then x_j is not invertible and hence belongs to the maximal ideal $\mathfrak{m}A_j$ of A_j; therefore $x \in \mathfrak{m}A'$.

The monogenic case treated above now gives a local, flat A'-algebra A_i such that

$$A_i/\mathfrak{m}A_i \simeq k_{<i}(t_i) = k_i.$$

The algebra A_i is flat over every A_j, $j < i$, so $i \in J$, contrary to the choice of i. This contradiction proves that $J = I$, and A_ξ has the required properties. This proves Lemma 4.1.2.

Let us now prove Theorem 4.1(iii). Recall once more that $\underline{P}$ denotes the set underlying a scheme P, and that $\underline{d} : \underline{P} \to \underline{Q}$ is the map induced by a morphism $d : P \to Q$. Part b) can then be restated by saying that the map

$$\underline{d_0} \boxtimes \underline{d_1} : \underline{X_1} \longrightarrow \underline{X_0} \times_{\underline{Y}} \underline{X_0}$$

with components $\underline{d_0}$ and $\underline{d_1}$ is surjective. This map factors as

$$\underline{X_1} \xrightarrow{d_0 \boxtimes d_1} \underline{X_0} \times_{\underline{Y}} \underline{X_0} \xrightarrow{q} \underline{X_0 \times_Y X_0},$$

where q is the canonical map. The image of $\underline{d_0} \boxtimes \underline{d_1}$ therefore contains every point $v \in \underline{X_0 \times_Y X_0}$ such that

$$\{v\} = q^{-1}(q(v)).$$

This condition is satisfied in particular when v is *rational over* Y, that is, when the residue field $\kappa(v)$ is identified with the residue field $\kappa(w)$ of the image w of v in Y.[23]

[23]Editor's note: the pages of Lecture Notes in Mathematics 151 are permuted here; their actual order is 265–266–268–269–267–270–271.

If $v \in \underline{X_0 \times_Y X_0}$ is not rational over Y, again let w be its image in Y. By Lemma 4.1.2, there exist a local ring C with radical $\mathfrak{m}$ and a local, flat homomorphism

$$f : \mathcal{O}_{Y,w} \longrightarrow C$$

such that $C/\mathfrak{m}$ is isomorphic to $\kappa(v)$ as a $\kappa(w)$-algebra. Put $Y' = \operatorname{Spec} C$, and let $\pi : Y' \to Y$ be the morphism induced by f. The canonical projection

$$(X_0 \times_Y X_0) \times_Y Y' \longrightarrow X_0 \times_Y X_0$$

sends some point v', rational over Y', to v. Since

$$(X_0 \times_Y X_0) \times_Y Y' \simeq (X_0 \times_Y Y') \times_{Y'} (X_0 \times_Y Y'),$$

and since the hypotheses of Theorem 4.1 and the preceding results, in particular part b), remain valid after the base change $\pi : Y' \to Y$, the point v' is the image of an element

$$u' \in \underline{X_1 \times_Y Y'}$$

under the base change of $d_0 \boxtimes d_1$. If u is the image of u' in X_1, then

$$v = (d_0 \boxtimes d_1)(u),$$

as required.

f) Proof of (iv).[24]

Lemma 4.1.3.— *If a monomorphism of schemes $f : T \to Z$ is finite, it is a closed immersion.*

Covering Z by affine open subsets Z_i, and replacing f by the induced morphisms $f^{-1}(Z_i) \to Z_i$, reduces us—since a finite morphism is affine—to the case

$$Z = \operatorname{Spec} B, \qquad T = \operatorname{Spec} A.$$

Because f is a monomorphism, the diagonal morphism

$$T \longrightarrow T \times_Z T$$

is an isomorphism (EGA I, 5.3.8); equivalently, multiplication gives an isomorphism

$$A \otimes_B A \xrightarrow{\sim} A.$$

Consequently, for every maximal ideal $\mathfrak{m}$ of B, there is an isomorphism

$$(A/\mathfrak{m}A) \otimes_k (A/\mathfrak{m}A) \xrightarrow{\sim} A/\mathfrak{m}A, \qquad k = B/\mathfrak{m}.$$

Since A is finite over B, the k-vector space $A/\mathfrak{m}A$ has finite dimension d, and this isomorphism implies $d = 0$ or $d = 1$. Hence the map

$$k = B/\mathfrak{m} \longrightarrow A/\mathfrak{m}A$$

is surjective. Nakayama's lemma, applied after localization at $\mathfrak{m}$, shows that $B_\mathfrak{m} \to A_\mathfrak{m}$ is surjective. It follows that $B \to A$ is surjective: its cokernel C satisfies $C_\mathfrak{m} = 0$ for every maximal ideal $\mathfrak{m}$, and therefore vanishes. This proves the lemma.

[24]Editor's note: the following lemma has been added from [DG70], I, §5, Proposition 1.5; see also the proof of EGA IV$_3$, 8.11.5. It was used implicitly in the original and explicitly in [DG70], III, §2, 4.6. It is also useful in Theorem 7.1 below.

Let us now prove (iv). By hypothesis,

$$X_0 = \operatorname{Spec} A_0, \qquad X_1 = \operatorname{Spec} A_1,$$

and, for $i = 0, 1$, the homomorphism $\delta_i : A_0 \to A_1$ makes A_1 a finite A_0-module. Therefore, a fortiori, the homomorphism

$$A_0 \otimes_B A_0 \longrightarrow A_1$$

is finite.

We assume in addition that

$$d = d_0 \boxtimes d_1 : X_1 \longrightarrow X_0 \times_Y X_0$$

is a monomorphism. By Lemma 4.1.3, the corresponding homomorphism $A_0 \otimes_B A_0 \to A_1$ is surjective. We shall prove that it is an isomorphism; in the course of doing so, we shall also prove that $p : X_0 \to Y$ is finite locally free.

It is enough to prove that, for every prime ideal $\mathfrak{p}$ of B, the homomorphism

$$(A_0)_\mathfrak{p} \otimes_{B_\mathfrak{p}} (A_0)_\mathfrak{p} \longrightarrow (A_1)_\mathfrak{p},$$

with components $\delta_{0\mathfrak{p}}$ and $\delta_{1\mathfrak{p}}$, is bijective. Thus we may suppose that B is local. Part b) then shows that $(A_0)_\mathfrak{p}$ is semilocal. Indeed, if $\mathfrak{m}$ is one of its maximal ideals, the others have the form $\delta_0^{-1}(\mathfrak{n})$, where $\mathfrak{n}$ ranges over the prime ideals of A_1 satisfying $\delta_1^{-1}(\mathfrak{n}) = \mathfrak{m}$. There are at most

$$n = [A_1 : A_0]$$

such prime ideals $\mathfrak{n}$.

After a faithfully flat base change, we may also suppose that the residue field of B is infinite.[25] We can then use the following lemma.

Lemma 4.2.— *Let B be a local ring with infinite residue field, let A be a semilocal ring, and let $i : B \to A$ be a homomorphism which sends the maximal ideal $\mathfrak{n}$ of B into the radical $\mathfrak{r}$ of A. Let M be a free A-module of rank n, and let N be a B-submodule of M which generates M as an A-module. Then N contains an A-basis of M.*

Recall that a sequence $m_1, \dots, m_n$ of elements of M is an A-basis if and only if its canonical images in $M/\mathfrak{r}M$ form a basis over $A/\mathfrak{r}$. We may therefore replace

$$M \text{ by } M/\mathfrak{r}M, \quad N \text{ by } N/(N \cap \mathfrak{r}M), \quad A \text{ by } A/\mathfrak{r}, \quad B \text{ by } B/\mathfrak{n}.$$

In this case the lemma is elementary. Namely, if

$$A = K_1 \times \cdots \times K_r, \qquad M = K_1^n \times \cdots \times K_r^n,$$

choose, for each j, an element $x_j \in N$ whose j-th component is nonzero. Because the field B is infinite, a suitable B-linear combination x of the x_j has all its components nonzero. Replace M by M/Ax, and proceed by induction on n.

Apply the lemma with $A = A_0$, with $i : B \hookrightarrow A_0$, with $M = A_1$ regarded as an A_0-module through δ_1, and with $N = \delta_0(A_0)$. Indeed, since

$$d_0 \boxtimes d_1 : X_1 \longrightarrow X_0 \times_Y X_0$$

is a closed immersion, the homomorphism

$$A_0 \otimes_B A_0 \longrightarrow A_1$$

[25] Editor's note: see Lemma 4.1.2.

with components δ_0, δ_1 is surjective. This says exactly that $\delta_0(A_0)$ generates the A_0-module A_1.

Choose $a_1, \dots, a_n \in A_0$ so that $\delta_0(a_1), \dots, \delta_0(a_n)$ form an A_0-basis of A_1. If $a_1, \dots, a_n$ form a B-basis of A_0, then $A_0 \otimes_B A_0 \to A_1$ sends the basis

$$(1 \otimes a_i)_{1 \le i \le n}$$

to the basis

$$(\delta_0(a_i))_{1 \le i \le n},$$

and is therefore bijective. Thus, if

$$\varepsilon : \mathbb{Z}^n \longrightarrow A_0$$

is the homomorphism of abelian groups which sends the standard basis of $\mathbb{Z}^n$ to $a_1, \dots, a_n$, it remains only to prove that the map

$$B \otimes_{\mathbb{Z}} \mathbb{Z}^n \longrightarrow A_0$$

with components i and ε is bijective.

The dual diagram $(0, 1, 2)^*$ considered at the beginning of this proof induces the following commutative diagram:

$$\begin{tikzcd}
A_2 & & A_1 \arrow[ll, shift right, "\delta'_1"'] \arrow[ll, shift left, "\delta'_0"'] & & A_0 \arrow[ll, "\delta_0"] \\
A_1 \otimes_{\mathbb{Z}} \mathbb{Z}^n \arrow[u, "u_2"] & & A_0 \otimes_{\mathbb{Z}} \mathbb{Z}^n \arrow[u, "u_1", "\cong"'] \arrow[ll, shift right, "\delta_1 \otimes \mathbb{Z}^n"'] \arrow[ll, shift left, "\delta_0 \otimes \mathbb{Z}^n"'] & & B \otimes_{\mathbb{Z}} \mathbb{Z}^n \arrow[u, "u_0"] \arrow[ll, "i \otimes \mathbb{Z}^n"]
\end{tikzcd}$$

Figure 22: The commutative ring-and-module diagram concluding the affine proof of Theorem 4.1(iv).

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Polo–Gille Exp5-13oct24.pdf, component-local p. 17; combined-reader p. 279; re-edited page header p. 267.

Here u_0, u_1, u_2 have, respectively, the component pairs

$$(i, \varepsilon), \qquad (\delta_1, \delta_0\varepsilon), \qquad (\delta'_2, \delta'_0\delta_0\varepsilon).$$

We know that u_1 is an isomorphism. Since the two left-hand squares of $(0, 1, 2)^*$ are cocartesian, u_2 is bijective. The two horizontal rows of the diagram are exact, and therefore u_0 is bijective.[26] Thus A_0 is a locally free B-module of rank n. By the preceding reductions, it follows that

$$\delta_0 \otimes \delta_1 : A_0 \otimes_B A_0 \xrightarrow{\sim} A_1$$

is an isomorphism. This completes the proof of Theorem 4.1 in the special case under consideration: X_0 is affine and d_1 is locally free of constant rank n.

[26]Editor's note: the argument which follows has been added.

5. Passage to the quotient by a finite flat groupoid (the general case)

a) Let $U^{(n)}$ be the largest open subset of X_0 over which d_1 is finite locally free of rank n. The scheme X_0 is the disjoint union of the $U^{(n)}$. The two Cartesian squares

$$\begin{tikzcd} X_2 \arrow[r, "d_0'"] \arrow[d, "d_2'"'] & X_1 \arrow[d, "d_1"] \\ X_1 \arrow[r, "d_0"'] & X_0 \end{tikzcd} \quad \textit{and} \quad \begin{tikzcd} X_2 \arrow[r, "d_1'"] \arrow[d, "d_2'"'] & X_1 \arrow[d, "d_1"] \\ X_1 \arrow[r, "d_1"'] & X_0 \end{tikzcd}$$

Figure 23: The two Cartesian squares controlling the rank strata $U^{(n)}$.

Native Loop-2 reconstructions; source locator: Polo–Gille Exp5-13oct24.pdf, component-local p. 17; combined-reader p. 279; re-edited page header p. 267.

show that the inverse images of $U^{(n)}$ under d_0 and d_1 both coincide with the largest open subset of X_1 over which d_2' is locally free of rank n.[27] Thus

$$d_0^{-1}(U^{(n)}) = d_1^{-1}(U^{(n)}),$$

and the groupoid X_* is the disjoint union of the groupoids $X_*^{(n)}$ induced by X_* on the open-and-closed subschemes $U^{(n)}$. It is therefore enough to prove Theorem 4.1 for each $X_*^{(n)}$. We are reduced to the case in which d_1 is finite locally free of rank n.

b) *We can now prove the theorem in the general case.*

By part a), we may suppose that d_1 is locally free of rank n. Let (Y, p) be a cokernel of (d_0, d_1) in the category of all ringed spaces. The argument at the end of 4(c) shows that, in order to prove Theorem 4.1(i), it is enough to show that Y is a scheme and that $p : X_0 \to Y$ is a morphism of schemes. By Lemma 1.2, this question is local on Y. Let $y \in Y$ and choose $x \in X_0$ with $p(x) = y$. If x has a saturated affine open neighborhood U, then Section 4 shows that $p(U)$ is an affine open subset of Y, and $p|_U : U \to p(U)$ is a morphism of schemes. Thus it remains to prove that every $x \in X_0$ has a saturated affine open neighborhood. We proceed as follows; the argument is taken from SGA 1, VIII, Corollary 7.6.

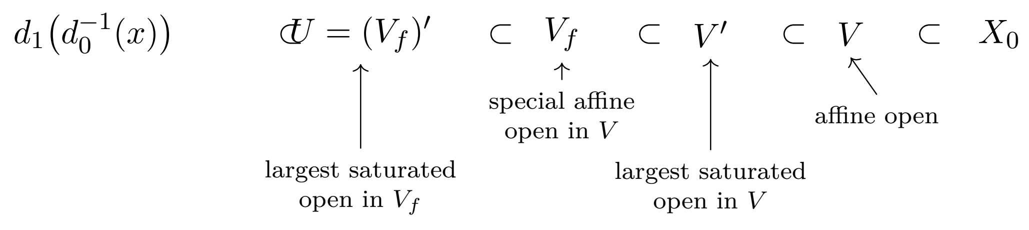

Figure 24: The nested open subsets used to construct a saturated affine neighborhood of x.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Polo–Gille Exp5-13oct24.pdf, component-local p. 18; combined-reader p. 280; re-edited page header p. 268.

[27] Editor's note: since d_0 (respectively d_1) is surjective, flat, and finite, hence faithfully flat and affine, d_2' has rank n over a neighborhood of a point $x \in X_1$ if and only if d_1 has rank n over a neighborhood of $d_0(x)$ (respectively $d_1(x)$).

By condition 4.1(b), there is an affine open subset $V \subseteq X_0$ which contains $d_1(d_0^{-1}(x))$.[28] Put $F = X_0 \setminus V$. Since d_1 is integral, $d_1(d_0^{-1}(F))$ is closed, and

$$V' = X_0 \setminus d_1(d_0^{-1}(F))$$

is the largest saturated open subset contained in V. The set V' is a neighborhood of the finite set $d_1(d_0^{-1}(x))$. Hence there is a section f of the structure sheaf of V, vanishing on $V \setminus V'$, such that $d_1(d_0^{-1}(x))$ is contained in the principal open subset $V_f \subseteq V$. We shall prove that the largest saturated open subset $(V_f)'$ contained in V_f is affine; it will then be the required neighborhood.

Put $Z(f) = V' \setminus V_f$. Then $d_0^{-1}(Z(f))$ is the set of points of

$$d_0^{-1}(V') = d_1^{-1}(V')$$

at which $d_0^*(f)$ vanishes. The morphism induced by d_1 from this common inverse image to V' is locally free of rank n.[29] Lemma 4.1.1 therefore shows that $d_1(d_0^{-1}(Z(f)))$ is precisely the zero locus of the norm N of $d_0^*(f)$ relative to d_1. It follows that

$$(V_f)' = V' \setminus d_1(d_0^{-1}(Z(f)))$$

is the set of points of V_f at which N does not vanish. Consequently, $(V_f)'$ is affine.

This proves Theorem 4.1(i); assertions (ii), (iii), and the first part of (iv) are then clear. It remains to prove the consequences stated at the end of (iv); see [Ray67a], Theorem 1(iii).

By hypothesis, the groupoid X_* comes from an equivalence relation

$$i : R \longrightarrow X_0 \times X_0,$$

where i is an immersion (see editor's note 19). We have shown that R is effective (see Exposé IV, 3.3.2), and that

$$p : X_0 \longrightarrow Y = X_0/R$$

is surjective and finite locally free; in particular, it is faithfully flat and of finite presentation.

Consequently, if (M) denotes the family of faithfully flat morphisms locally of finite presentation, then R is (M)-effective. By Exposé IV, 6.3.3, (Y, p) therefore represents the quotient sheaf of X_0 by R for the fppf topology, and the base-change assertions follow from Exposé IV, 3.4.3.1.

Remark 5.1.—[30] Keep the hypotheses and notation of Theorem 4.1, and suppose in addition that S is locally Noetherian and that $\pi_0 : X_0 \to S$ is quasi-projective. We shall prove that $\pi : Y \to S$ is quasi-projective.

The preceding hypotheses imply that $Y \to S$ is of finite type; see the proof of 6.1(ii). Let $\mathscr{A}$ be an invertible $\mathcal{O}_{X_0}$-module which is ample for π_0. By EGA II, 6.1.12, $p_*\mathscr{A}$ is an invertible $p_*\mathcal{O}_{X_0}$-module. Hence Y has an affine open cover $(V_i)_{i \in I}$ such that $\mathscr{A}$ is trivial over each of the saturated affine open subsets

$$U_i = p^{-1}(V_i).$$

For each i, put

$$A_{i,0} = \mathcal{O}_{X_0}(U_i),$$

[28] Editor's note: $d_1(d_0^{-1}(x)) = d_0(d_1^{-1}(x))$; see editor's note 16 in Theorem 4.1.
[29] Editor's note: the reference to Lemma 4.1.1 has been added; see [DG70], III, §5.2, p. 313.
[30] Editor's note: this paragraph has been added.

and let $A_{i,1}$ be the ring of the affine open subset $d_0^{-1}(U_i) = d_1^{-1}(U_i)$ of X_1. Let $\delta_{i,0}$ (respectively $\delta_{i,1}$) be the homomorphism $A_{i,0} \to A_{i,1}$ induced by d_0 (respectively d_1), and put

$$B_i = \mathcal{O}_Y(V_i) = \{b \in A_{i,0} \mid \delta_{i,0}(b) = \delta_{i,1}(b)\}.$$

Following EGA II, §6.5, consider the invertible $\mathcal{O}_{X_0}$-module

$$N_{d_1}(d_0^*\mathscr{A}),$$

the norm, relative to the finite locally free morphism $d_1 : X_1 \to X_0$, of the invertible $\mathcal{O}_{X_1}$-module $d_0^*\mathscr{A}$. Suppose that $\mathscr{A}$, relative to the cover $(U_i)_{i\in I}$, is given by transition functions

$$c_{ij} \in \mathcal{O}_{X_0}(U_i \cap U_j)^\times.$$

Then $N_{d_1}(d_0^*\mathscr{A})$ is given by the transition functions

$$N_{\delta_1}(\delta_0(c_{ij})) \in \mathcal{O}_{X_0}(U_i \cap U_j)^\times.$$

By 4(a), these elements belong to $\mathcal{O}_Y(V_i \cap V_j)^\times$, and hence define an invertible $\mathcal{O}_Y$-module $\mathscr{L}$ such that

$$p^*\mathscr{L} = N_{d_1}(d_0^*\mathscr{A}).$$

Moreover, for every $n \in \mathbb{N}^*$,

$$p^*(\mathscr{L}^n) = N_{d_1}(d_0^*(\mathscr{A}^n));$$

see *loc. cit.*, (6.5.2.1).

We now show that $\mathscr{L}$ is ample for $\pi : Y \to S$. Replacing S by an affine open subset, we may suppose that S is affine. Let $y \in Y$, choose $x \in X_0$ with $p(x) = y$, let V be an affine open neighborhood of y in Y, and put $U = p^{-1}(V)$. Since $\mathscr{A}$ is π_0-ample, there exist $n \in \mathbb{N}^*$ and a section

$$s \in \Gamma(X_0, \mathscr{A}^n)$$

such that

$$x \in (X_0)_s \subseteq U.$$

With the notation above, s is represented by sections

$$a_i \in A_{i,0} = \mathcal{O}_{X_0}(U_i), \qquad a_i = c_{ij}a_j \ \text{ on } \ U_i \cap U_j,$$

and $(X_0)_s$ is the union of the open subsets

$$U_i' = \{\mathfrak{p} \in \operatorname{Spec}(A_{i,0}) \mid a_i \notin \mathfrak{p}\}.$$

For every i, put

$$N(a_i) = N_{\delta_1}(\delta_0(a_i)) \in B_i.$$

Theorem 4.1(i) and Lemma 4.1.1 give

$$p(U_i') = p\, d_1 d_0^{-1}(U_i') = p\, d_1(\{\mathfrak{q} \in \operatorname{Spec}(A_{i,1}) \mid \delta_{i,0}(a_i) \notin \mathfrak{q}\}),$$

while

$$d_1(\{\mathfrak{q} \in \operatorname{Spec}(A_{i,1}) \mid \delta_{i,0}(a_i) \notin \mathfrak{q}\}) = \{\mathfrak{p} \in \operatorname{Spec}(A_{i,0}) \mid N_{\delta_1}(\delta_{i,0}(a_i)) \notin \mathfrak{p}\}.$$

It follows that

$$p(U_i') = \{\mathfrak{p} \in \operatorname{Spec}(B_i) \mid N(a_i) \notin \mathfrak{p}\}.$$

Consequently,

$$p((X_0)_s) = Y_{N(s)},$$

where $N(s)$ is the section of $\mathscr{L}^n$ on Y defined by the sections $N(a_i) \in \mathcal{O}_Y(V_i)$. Thus

$$y \in p((X_0)_s) = Y_{N(s)} \subseteq p(U) = V. \tag{*}$$

This proves that $\mathscr{L}$ is ample for $\pi : Y \to S$, and hence completes the proof that π is quasi-projective.

6. Passage to the quotient when a quasi-section exists

We shall now prove a technical lemma that will be used in the proof of the two theorems we have in view. Let S be a scheme and let

$$X_2 \overset{d'_0, d'_1, d'_2}{\rightrightarrows\!\!\!\to} X_1 \overset{d_0, d_1}{\rightrightarrows} X_0$$

Figure 25: The groupoid X_*.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Exposé V, component-local p. 20 / combined-reader p. 282 / re-edited header p. 270.

be a $(\mathbf{Sch}/S)$-groupoid. We call a *quasi-section* of the groupoid X_* any subscheme U of X_0 for which the following two conditions hold:

(1) The restriction v of d_1 to $d_0^{-1}(U)$ is a finite, locally free, surjective morphism from $d_0^{-1}(U)$ onto X_0.

(2) Every subset E of U consisting of points that are pairwise equivalent for the equivalence relation defined by X_* (§3.e)) is contained in an affine open subset of U.[31]

If U is a quasi-section of X_*, the $(\mathbf{Sch}/S)$-groupoid

$$U_2 \overset{u'_0, u'_1, u'_2}{\rightrightarrows\!\!\!\to} U_1 \overset{u_0, u_1}{\rightrightarrows} U$$

Figure 26: The groupoid U_* induced by X_*.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Exposé V, component-local p. 20 / combined-reader p. 282 / re-edited header p. 270.

induced by X_* and the inclusion $U \hookrightarrow X_0$ (see §3.a)) satisfies the hypotheses of Theorem 4.1. Indeed, put

$$V = d_0^{-1}(U),$$

and let u and v be the morphisms with source V induced by d_0 and d_1, respectively:

$$X_0 \xleftarrow[v]{} V \xrightarrow{u} U$$

Figure 27: The morphisms $v : V \to X_0$ and $u : V \to U$.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Exposé V, component-local p. 20 / combined-reader p. 282 / re-edited header p. 270.

By §3.b), there is a Cartesian square

[31] Editor's note: If $x, y \in E$, there is a $z \in X_1$ such that $d_1(z) = x$ and $d_0(z) = y$; thus z belongs to $v^{-1}(x)$, which is finite by (1). Hence E is contained in the finite set $d_0(v^{-1}(x)) = d_0 d_1^{-1}(x) \cap U$. The source PDF prints $d_1(x) = x$ in the first equality; the type-correct reading $d_1(z) = x$ is used here.

$$\begin{tikzcd} U_1 \arrow[r] \arrow[d, "u_1"'] & V \arrow[d, "v"] \\ U \arrow[r, "\text{inclusion}"'] & X_0 \end{tikzcd}$$

Figure 28: The Cartesian square defining U_1.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Exposé V, component-local p. 20 / combined-reader p. 282 / re-edited header p. 270.

so u_1 is surjective and finite locally free by condition (1). Together with condition (2), condition (1) therefore ensures that the groupoid U_* satisfies the hypotheses of Theorem 4.1. In particular, $\operatorname{Coker}(u_0, u_1)$ exists in $(\mathbf{Sch}/S)$. Moreover, d_0 has a section, so u is a universal effective epimorphism (see Exposé III, 1.12). Proposition 3.1 then shows that $\operatorname{Coker}(u_0, u_1)$ coincides with the cokernel $\operatorname{Coker}(v_0, v_1)$ of the groupoid V_*:

$$\begin{tikzcd} V_2 \arrow[r, shift left=2, "v_2'"] \arrow[r, "v_1'"] \arrow[r, shift right=2, "v_0'"'] & V_1 \arrow[r, shift left, "v_1"] \arrow[r, shift right, "v_0"'] & V \end{tikzcd}$$

Figure 29: The groupoid V_*.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Exposé V, component-local p. 20 / combined-reader p. 282 / re-edited header p. 270.

This is the inverse image of U_* under the base change $u : V \to U$, and hence also the inverse image of X_* under the base change

$$V \xrightarrow{\text{inclusion}} X_1 \xrightarrow{d_0} X_0$$

Figure 30: The base change defining V_*.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Exposé V, component-local p. 21 / combined-reader p. 283 / re-edited header p. 271.

By §3.c), V_* is isomorphic to the groupoid V'_*, the inverse image of X_* under the base change

$$v\colon \qquad V \xrightarrow{\text{inclusion}} X_1 \xrightarrow{d_1} X_0$$

Figure 31: The base change defining V'_*.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Exposé V, component-local p. 21 / combined-reader p. 283 / re-edited header p. 271.

Thus V'_* admits a cokernel in $(\mathbf{Sch}/S)$. Since $v : V \to X_0$ is flat, surjective, and finite, it is faithfully flat and quasi-compact, and therefore a universal effective epimorphism by Exposé III, 6.3.2. Proposition 3.1 consequently shows that X_* admits a cokernel $\operatorname{Coker}(d_0, d_1)$ in $(\mathbf{Sch}/S)$. We have therefore proved the first assertion of part (i) of the following lemma.[32]

[32]Editor's note: The argument that follows has been slightly modified. In particular, the additional

Lemma 6.1.—*Suppose that the* $(\mathbf{Sch}/S)$*-groupoid* X_* *possesses a quasi-section. Then:*

(i) *There exists a cokernel* (Y, p) *of* (d_0, d_1) *in* $(\mathbf{Sch}/S)$*. Moreover, every such* (Y, p) *is a cokernel of* (d_0, d_1) *in the category of all ringed spaces.*

(i') *The morphism p is surjective; it is open (respectively, universally closed) if* d_0 *is.*

(ii) *Suppose that* S *is locally Noetherian and that* X_0 *is locally of finite type (respectively, of finite type) over* S*. Then* p *and* $Y \to S$ *are locally of finite presentation (respectively, of finite presentation).*

(iii) *The morphism* $X_1 \to X_0 \times_Y X_0$ *with components* d_0 *and* d_1 *is surjective.*

(iv) *If* (d_0, d_1) *is an equivalence pair, then* $X_1 \to X_0 \times_Y X_0$ *is an isomorphism. Moreover, if* $d_0 : X_1 \to X_0$ *is flat, then* p *is faithfully flat.*

Before proving the second assertion of part (i), we shall prove parts (i'), (ii), and (iii).

a) Proof of (i') and (ii).—We have just seen that (Y, p) identifies both with $\mathrm{Coker}(v_0, v_1)$ and with $\mathrm{Coker}(u_0, u_1)$. Let q and r be the canonical epimorphisms from U and V, respectively, to Y:

$$\begin{tikzcd}
X_0 \arrow[dr, "p"'] & V \arrow[l, "v"] \arrow[r, "u"] \arrow[d, "r"] & U \arrow[dl, "q"] \\
& Y &
\end{tikzcd}$$

Figure 32: The canonical epimorphisms to Y.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Exposé V, component-local p. 21 / combined-reader p. 283 / re-edited header p. 271.

By hypothesis, v is surjective and finite locally free, hence open.

On the other hand, if $d_0 : X_1 \to X_0$ is open (respectively, universally closed), then u, being obtained from d_0 by base change, has the same property. By Theorem 4.1, q is surjective, integral, and open. It follows that r is surjective and is open (respectively, universally closed) if d_0 is. Since v is surjective, the same is true of p. This proves part (i').

Now suppose that S is locally Noetherian and that X_0 is locally of finite type over S, so that X_0 is locally Noetherian.

We show that Y is locally of finite presentation over S. Let

$$S' = \operatorname{Spec} R$$

be an affine open subset of S, let

$$Y' = \operatorname{Spec} B$$

be an affine open subset of Y mapping into S', and let

$$U' = \operatorname{Spec} A$$

be the inverse image of Y' in U. Since R is Noetherian, it is enough to show that B is a finitely generated R-algebra. By §§4 and 5, B is contained in A, which is a finitely generated R-algebra; the assertion therefore follows because R is Noetherian and A is integral over B.

hypothesis that d_0 be flat has been moved to part (iv); the former part (ii) has been split into parts (i') and (ii); and the second assertion of part (i') has been added.

Finally, since $X_0 \to S$ is locally of finite type, so is p (EGA I, 6.6.6). Hence p is locally of finite presentation because Y is locally Noetherian.

It remains to prove the final assertion of part (ii). Suppose in addition that X_0 is of finite type over S. Since p is surjective, Y is then quasi-compact over S, and hence of finite type over S. Because S is locally Noetherian, the morphisms $X_0 \to S$ and $Y \to S$ are of finite presentation, and therefore so is $p : X_0 \to Y$ (EGA IV_1, 1.6.2(v)).

b) Proof of (iii).—The groupoid V_* with base V is isomorphic both to the inverse image of U_* under the base change u and to the inverse image of X_* under the base change v. We therefore have a double Cartesian square:

$$\begin{tikzcd}
X_1 \arrow[d, "d_0 \boxtimes d_1"'] & V_1 \arrow[l] \arrow[r] \arrow[d, "v_0 \boxtimes v_1"'] & U_1 \arrow[d, "u_0 \boxtimes u_1"'] \\
X_0 \times_Y X_0 & V \times_Y V \arrow[l, "v \times v"'] \arrow[r, "u \times u"] & U \times_Y U
\end{tikzcd}$$

Figure 33: The double Cartesian square used in the proof of Lemma 6.1(iii).

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Exposé V, component-local p. 22 / combined-reader p. 284 / re-edited header p. 272.

Since $u_0 \boxtimes u_1$ is surjective, so is $v_0 \boxtimes v_1$. Since $v \times v$ is surjective, the composite morphism

$$V_1 \longrightarrow X_0 \times_Y X_0$$

is surjective, and hence so is $d_0 \boxtimes d_1$.

c) Proof of (i).—It remains to prove that (Y, p) is a cokernel of (d_0, d_1) in the category of all ringed spaces. We first show that Y is obtained from X_0 by identifying those points x and y for which there exists $z \in X_1$ with

$$d_0(z) = x, \qquad d_1(z) = y.$$

Indeed, p is surjective and $pd_0 = pd_1$. Conversely, if $p(x) = p(y)$, there is a point $z' \in X_0 \times_Y X_0$ whose first projection is x and whose second projection is y. If $z \in X_1$ satisfies

$$(d_0 \boxtimes d_1)(z) = z',$$

then $d_0(z) = x$ and $d_1(z) = y$, as required.

On the other hand, if W is a saturated open subset of X_0, then $W \cap U$ is a saturated open subset of U. By Theorem 4.1, $q(W \cap U)$ is open in Y. Since $q(W \cap U) = p(W)$, the topology on Y is the quotient topology induced by X_0.

It remains to show that the canonical sequence of sheaves of rings

$$\mathcal{O}_Y \longrightarrow p_*\mathcal{O}_{X_0} \rightrightarrows p_*d_{0*}\mathcal{O}_{X_1} = p_*d_{1*}\mathcal{O}_{X_1} \tag{4}$$

is exact.

Let Y' be an open subset of Y, and put

$$U' = q^{-1}(Y'), \qquad X_0' = p^{-1}(Y'),$$

and so forth.[33] Then U' is an open subset of U saturated for the equivalence relation defined by the groupoid U_*. It follows from Lemmas 1.1 and 1.2 that Y' is the cokernel,

[33] Editor's note: The argument that follows has been expanded, in particular the verification that U' is a quasi-section of the groupoid induced on X_0'.

both in $(\mathbf{Sch}/S)$ and in the category of ringed spaces, of the groupoid induced by U_* on U'. Likewise, X'_0 is an open subset of X_0 saturated for the equivalence relation defined by X_*, and we have the following commutative diagram, in which both squares are Cartesian:

$$\begin{tikzcd}
X'_0 & V' = d_0^{-1}(U') \arrow[l, "\widetilde{d}_1"'] \arrow[r, "\widetilde{d}_0"] \arrow[d] & U' \arrow[d] \\
X_0 & V = d_0^{-1}(U) \arrow[l, "d_1"'] \arrow[r, "d_0"'] & U
\end{tikzcd}$$

Figure 34: The restricted groupoids over X'_0 and X_0.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Exposé V, component-local p. 23 / combined-reader p. 285 / re-edited header p. 273; lower-right U corrected from source-PDF U'.

Here $\widetilde{d}_1$ is surjective and finite locally free. On the other hand, let $x \in U'$. Since U is a quasi-section, the set

$$E = d_0 d_1^{-1}(x) \cap U$$

is finite and is contained in an affine open subset W of U. Then $E' = E \cap U'$ is a finite set contained in the quasi-affine open subset $W \cap U'$. Consequently there is an affine open subset W' of $W \cap U'$ that contains E'. This proves that U' is a quasi-section of the groupoid X'_* induced by X_* on X'_0. The first assertion of part (i), applied to X'_* and U', now shows that Y' is the cokernel of X'_* in $(\mathbf{Sch}/S)$.

In particular, for every S-scheme T, the sequence

$$T(Y') \xrightarrow{T(p|_{X'_0})} T(X'_0) \underset{T(d_0|_{X'_1})}{\overset{T(d_1|_{X'_1})}{\rightrightarrows}} T(X'_1) \tag{5}$$

is exact. If T is the affine line $\mathbf{G}_{a,S}$ (Exposé I, 4.3), this sequence identifies with

$$\begin{aligned}\Gamma(Y', \mathcal{O}_Y) \longrightarrow \Gamma(p^{-1}(Y'), \mathcal{O}_{X_0}) &\underset{\delta_0}{\overset{\delta_1}{\rightrightarrows}} \Gamma(d_0^{-1}p^{-1}(Y'), \mathcal{O}_{X_1}) \\ &= \Gamma(d_1^{-1}p^{-1}(Y'), \mathcal{O}_{X_1}),\end{aligned} \tag{6}$$

which is therefore exact for every open subset Y'. This completes the proof of Lemma 6.1(i).

d) Proof of (iv).—If (d_0, d_1) is an equivalence pair, then so is (u_0, u_1). It follows from Theorem 4.1 that

$$u_0 \boxtimes u_1 : U_1 \longrightarrow U \times_Y U$$

is an isomorphism, and hence so is $v_0 \boxtimes v_1$; see the Cartesian squares in part (b). Since $v \times v$ is faithfully flat and quasi-compact, $d_0 \boxtimes d_1$ is an isomorphism (SGA 1, VIII, 5.4).

Moreover, if d_0 is flat, then u is flat. The morphism q is flat by Theorem 4.1, so r is flat as well. Since v is faithfully flat, p is flat; being also surjective, it is faithfully flat.

7. Quotient by a proper and flat groupoid

Theorem 7.1.—[34]*Let S be a locally Noetherian scheme and let*

[34]Editor's note: We mention here the article of S. Keel and S. Mori [KM97], which establishes the following theorem. Let X be an algebraic space of finite type over a locally Noetherian base S, and let $j : R \to X \times_S X$

$$
\begin{tikzcd}
X_2 \arrow[r, shift left=2, "d'_2"] \arrow[r, "d'_1"] \arrow[r, shift right=2, "d'_0"'] & X_1 \arrow[r, shift left, "d_1"] \arrow[r, shift right, "d_0"'] & X_0
\end{tikzcd}
$$

Figure 35: The groupoid occurring in Theorem 7.1.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Exposé V, component-local p. 24 / combined-reader p. 286 / re-edited header p. 274.

be a $(\mathbf{Sch}/S)$*-groupoid such that* d_1 *is proper and flat,* X_0 *is quasi-projective over* S*,*[35] *and the morphism*

$$d : X_1 \longrightarrow X_0 \times_S X_0$$

with components d_0 *and* d_1 *is quasi-finite. Then:*

(i) *There exists a cokernel* (Y, p) *of* (d_0, d_1) *in* $(\mathbf{Sch}/S)$*. Moreover, every such* (Y, p) *is a cokernel of* (d_0, d_1) *in the category of all ringed spaces.*

(ii) *The morphism* p *is surjective, open, proper, and of finite presentation, and* $Y \to S$ *is separated and of finite presentation.*[36]

(iii) *The morphism*

$$X_1 \longrightarrow X_0 \times_Y X_0$$

with components d_0 *and* d_1 *is surjective.*

(iv) *If* (d_0, d_1) *is an equivalence pair, then* $X_1 \to X_0 \times_Y X_0$ *is an isomorphism and* p *is faithfully flat.*[37] *Moreover,* (Y, p) *is a cokernel of* (d_0, d_1) *in the category of sheaves for the fppf topology, and, for every base change* $Y' \to Y$*, the scheme* Y' *is the cokernel of the groupoid* $X_* \times_Y Y'$ *obtained from* X_* *by the base change* $X_0 \times_Y Y' \to X_0$*.*

In particular, for every base change $S' \to S$*, the scheme*

$$Y' = Y \times_S S'$$

is the cokernel of the S'*-groupoid*

$$X'_* = X_* \times_S S'.$$

Thus, in this case, "formation of the quotient commutes with base change".

be a flat groupoid whose stabilizer $j^{-1}(\Delta_X)$ is finite over X. There then exists an algebraic space that is a geometric quotient of X by R and a uniform categorical quotient; moreover, if j is separated, this quotient is separated. In particular, if a flat S-group scheme G acts properly on X, with finite stabilizer (that is, if

$$G \times_S X \longrightarrow X \times_S X, \qquad (g, x) \longmapsto (x, g \cdot x)$$

is proper and the stabilizer of the diagonal is finite over X), then there exists a geometric quotient $X \to X/G$. For a reductive S-group scheme G, this is a result of J. Kollár [Ko97].

[35] Editor's note: This hypothesis on X_0 is necessary; see editor's note (17) to Theorem 4.1.

[36] Editor's note: TDTE III, Theorem 6.1, states that $Y \to S$ is quasi-projective if S is Noetherian. The editors did not see how to deduce this from the local existence of quasi-sections.

[37] Editor's note: The consequences that follow have been made explicit; see [Ray67a], Theorem 1(iii), and the end of the proof of the theorem. We also mention that another proof of Theorem 7.1 in the case of an equivalence relation, based on the existence of Hilbert schemes, is given in [AK80], Theorem 2.9; see also [BLR90], §8.2, Theorem 12. In that case it is furthermore shown that $Y \to S$ is quasi-projective.

Let (Y, p) be the cokernel of (d_0, d_1) in the category of all ringed spaces. Lemma 1.2 shows that, to prove part (i), it is enough to show that every point $z \in X_0$ has a saturated open neighborhood U_z such that, if $\widetilde{d_0}$ and $\widetilde{d_1}$ denote the restrictions of d_0 and d_1 to

$$d_0^{-1}(U_z) = d_1^{-1}(U_z),$$

and if (Q, q) denotes the cokernel of $(\widetilde{d_0}, \widetilde{d_1})$ in the category of ringed spaces, then Q is a scheme and q is a morphism of schemes.

By Lemma 6.1(i), it is therefore enough to show that every point $z \in X_0$ has a saturated open neighborhood U_z such that the groupoid induced by X_* on U_z has a quasi-section. We may even assume that z is closed in its fiber over S (we shall say that z is *closed relative to* S).[38] The existence of U_z follows from the lemmas below.

Lemma 7.2.—*Let T be an affine Noetherian scheme, let X, Y, Z be T-schemes of finite type, with X quasi-projective over T, and let*

$$\begin{tikzcd} Y \arrow[r, "u"] \arrow[d, "v"'] & X \arrow[d, dashed] \\ Z \arrow[r, dashed] & T \end{tikzcd}$$

Figure 36: The diagram in Lemma 7.2.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Exposé V, component-local p. 25 / combined-reader p. 287 / re-edited header p. 275.

be a diagram in $(\mathbf{Sch}/T)$. *Let z be a point of $v(Y)$ that is closed relative to T, and suppose that v is flat at the points of $v^{-1}(z)$. Then there exists a closed subscheme F of X such that*

$$u\big(u^{-1}(F) \cap v^{-1}(z)\big)$$

is finite and nonempty, and such that the restriction of v to $u^{-1}(F)$ is flat at the points of $v^{-1}(z)$.

Put $T = \operatorname{Spec} A$. We may suppose that

$$X = \operatorname{Proj} \mathscr{S},$$

where $\mathscr{S}$ is the symmetric algebra of a finitely generated A-module E.

If $u(v^{-1}(z))$ is finite, we may take $F = X$. Otherwise, let

$$y_1, \ldots, y_n$$

be the points of the fiber $v^{-1}(z)$ associated with its structure sheaf $\mathcal{O}_{v^{-1}(z)}$. Thus, if $\mathcal{O}_i$ denotes the local ring of $v^{-1}(z)$ at y_i, the maximal ideal of $\mathcal{O}_i$ consists of zero divisors. Let t be the image of z in T. The set $u(v^{-1}(z))$ is an infinite constructible subset of the fiber of X over t. Consequently, that fiber contains a closed point $x \in u(v^{-1}(z))$ distinct from

[38] Editor's note: Indeed, if such an open neighborhood U_z has been constructed for every point z that is closed relative to S, then the union of these U_z covers X_0: every fiber over S of the closed complement is a Noetherian scheme with no closed points, and is therefore empty.

$u(y_1), \dots, u(y_n)$. The open subset $X - \{x\}$ is a neighborhood of $u(y_1), \dots, u(y_n)$, and hence contains an open neighborhood of the form $D_+(f)$, where $f \in \mathscr{S}$ is homogeneous of degree d (we use the notation of EGA II, §2.3).

The closed subscheme

$$X_1 = V_+(f)$$

defined by f therefore contains x and avoids the points $u(y_1), \dots, u(y_n)$. Its inverse image

$$Y_1 = u^{-1}\big(V_+(f)\big)$$

is consequently distinct from Y and meets $v^{-1}(z)$. We shall in fact prove that the restriction v_1 of v to Y_1 is flat at the points of $v^{-1}(z)$. If $u(v_1^{-1}(z))$ is finite, we may then take $F = X_1$. Otherwise, we repeat the argument with Y_1, v_1, and the morphism u_1 induced by u on Y_1, in place of Y, v, u. In this way we obtain a descending sequence

$$X \supset X_1 \supset \cdots$$

of closed subschemes of X. Since such a sequence terminates,

$$u\big(u^{-1}(X_n) \cap v^{-1}(z)\big)$$

is finite and nonempty for some n, and we then take $F = X_n$.

It remains to show that v_1 is flat at the points of $v^{-1}(z)$. Let $y \in Y_1$ lie over z, let $\mathcal{O}_y$ be the local ring of Y at y, let $\overline{\mathcal{O}}_y$ be the local ring of $v^{-1}(z)$ at y, and let $\mathcal{O}_{v(y)}$ be the local ring of Z at $v(y)$. Choose $g \in \mathscr{S}_1$ such that $D_+(g)$ is a neighborhood of $u(y)$ in X, and let φ and $\overline{\varphi}$ be the respective images of f/g^d in $\mathcal{O}_y$ and $\overline{\mathcal{O}}_y$. By the construction of f, the element $\overline{\varphi}$ is not a zero divisor in $\overline{\mathcal{O}}_y$. Since $\mathcal{O}_y$ is flat over $\mathcal{O}_z = \mathcal{O}_{v(y)}$, the element φ is not a zero divisor in $\mathcal{O}_y$, and

$$\mathcal{O}_y/\mathcal{O}_y\varphi$$

is flat over $\mathcal{O}_z$ (SGA 1, IV, 5.7). This quotient is precisely the local ring of Y_1 at y.

Lemma 7.3.—*Retain the notation and hypotheses of Theorem 7.1. Every point $z \in X_0$ that is closed relative to S has a saturated open neighborhood U_z such that the groupoid induced by X_* on U_z has a quasi-section.*

The assertion is local on S, so we may suppose that S is affine and Noetherian and apply the preceding lemma to the diagram

$$\begin{tikzcd} X_1 \arrow[r, "d_0"] \arrow[d, "d_1"'] & X_0 \arrow[d, dashed] \\ X_0 \arrow[r, dashed] & S \end{tikzcd}$$

Figure 37: The diagram to which Lemma 7.2 is applied.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Exposé V, component-local p. 26 / combined-reader p. 288 / re-edited header p. 276.

in $(\mathbf{Sch}/S)$. Thus let F be a closed subscheme of X_0 such that

$$d_0\big(d_0^{-1}(F) \cap d_1^{-1}(z)\big)$$

is finite and nonempty, and such that the restriction of d_1 to $d_0^{-1}(F)$ is flat at the points of $d_1^{-1}(z)$.

Let F_1 and F_2 be the inverse images of F under d_0 and $d_0 d_0' = d_0 d_1'$, respectively, and let $\widetilde{d}_0, \widetilde{d}_1, \ldots$ denote the morphisms induced by $d_0, d_1, \ldots$. We then have a commutative diagram

$$\begin{tikzcd}
F_2 \arrow[r, shift left, "\widetilde{d}_1'"] \arrow[r, shift right, "\widetilde{d}_0'"'] \arrow[d, "\widetilde{d}_2'"'] & F_1 \arrow[r, "\widetilde{d}_0"] \arrow[d, "\widetilde{d}_1"] & F \arrow[d, dashed, "\widetilde{q}"] \\
X_1 \arrow[r, shift left, "d_1"] \arrow[r, shift right, "d_0"'] & X_0 \arrow[r, dashed, "q"] & S
\end{tikzcd}$$

Figure 38: The induced diagram on F_2, F_1, F.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Exposé V, component-local p. 26 / combined-reader p. 288 / re-edited header p. 276.

in which the two left-hand squares are Cartesian, the first row is exact (cf. the diagram $(0, 1, 2)$ in §1), and $q, \widetilde{q}$ are the structure morphisms.

We first show that only finitely many points of F_1 lie over z.[39] Let s be the image of z in S. Since F is of finite type over S, the fiber $\widetilde{q}^{-1}(s)$ is a Noetherian scheme. Since $\widetilde{d}_0$ is proper,

$$\widetilde{d}_0\big(\widetilde{d}_1^{-1}(z)\big)$$

is a closed subscheme of $\widetilde{q}^{-1}(s)$ consisting of finitely many points. Consequently (cf. EGA I, 6.2.2), those points are closed in $\widetilde{q}^{-1}(s)$, and also in the fiber $q^{-1}(s)$ of X_0, because F is closed in X_0.

Let y be one of these points. Since $q^{-1}(s)$ is of finite type over $\kappa(s)$, it has affine open neighborhoods

$$\operatorname{Spec} B \quad \text{and} \quad \operatorname{Spec} C$$

of y and z, respectively, with B and C finitely generated $\kappa(s)$-algebras. The points y and z correspond to maximal ideals

$$\mathfrak{p} \subset B, \qquad \mathfrak{q} \subset C.$$

The fields $B/\mathfrak{p}$ and $C/\mathfrak{q}$ are finite over $\kappa(s)$, so

$$(B/\mathfrak{p}) \otimes_{\kappa(s)} (C/\mathfrak{q})$$

is a finite-dimensional $\kappa(s)$-algebra. Its maximal ideals correspond precisely to the points of $X_0 \times_S X_0$ whose second projection is z and whose first projection is y. Thus there are only

[39] Editor's note: Details have been added, and the role of the properness hypothesis on d_0 and d_1 in Theorem 7.1 has been made explicit. Compare the statement and proof of Theorem 8.1, where this properness hypothesis is omitted.

finitely many points $u \in X_0 \times_S X_0$ whose second projection is z and whose first projection belongs to

$$\widetilde{d}_0\big(\widetilde{d}_1^{-1}(z)\big).$$

Finally, since $X_1 \to X_0 \times_S X_0$ has finite fibers, each such point u is the image of only finitely many points of X_1. This proves the assertion.

The morphism $\widetilde{d}_1$ is therefore quasi-finite and flat at the points of F_1 lying over z. Since $\widetilde{d}_1$ is of finite type, SGA 1, IV, 6.10 and EGA III, 4.4.10 imply that the set Φ of points of F_1 at which $\widetilde{d}_1$ is not both flat and quasi-finite is closed in F_1, and hence in X_1 because F_1 is closed in X_1.[40] Since d_1 is proper, $\widetilde{d}_1(\Phi)$ is closed and, by what precedes, does not contain z. Put

$$W = \widetilde{d}_1(F_1) - \widetilde{d}_1(\Phi).$$

The restriction of $\widetilde{d}_1$ to $\widetilde{d}_1^{-1}(W)$ is of finite presentation (by the Noetherian hypotheses), flat, proper, and quasi-finite.[41] It is therefore finite, locally free, and open, by EGA III, 4.4.2 and EGA IV$_2$, 2.1.12 and 2.4.6. Consequently, $\widetilde{d}_1(F_1)$ is a neighborhood of z, and W is the largest open subset of X_0 contained in $\widetilde{d}_1(F_1)$ over which $\widetilde{d}_1$ is both flat and quasi-finite.

We shall see in Lemma 7.4 that the inverse images of Φ under $\widetilde{d}'_1$ and $\widetilde{d}'_0$ both identify with the set of points of F_2 at which $\widetilde{d}'_2$ is not both flat and quasi-finite. It follows that

$$d_0^{-1}(W) = \widetilde{d}'_2(F_2) - \widetilde{d}'_2\big((\widetilde{d}'_0)^{-1}(\Phi)\big),$$
$$d_1^{-1}(W) = \widetilde{d}'_2(F_2) - \widetilde{d}'_2\big((\widetilde{d}'_1)^{-1}(\Phi)\big)$$

coincide. Thus W is saturated.

Set

$$W_1 = \widetilde{d}_1^{-1}(W).$$

The equality $d_0^{-1}(W) = d_1^{-1}(W)$ implies

$$(\widetilde{d}'_2)^{-1}\big(d_0^{-1}(W)\big) = (\widetilde{d}'_2)^{-1}\big(d_1^{-1}(W)\big),$$

that is,

$$(\widetilde{d}'_0)^{-1}(W_1) = (\widetilde{d}'_1)^{-1}(W_1).$$

The morphism $\widetilde{d}_0$ is faithfully flat and quasi-compact, because d_0, like d_1, is surjective, proper, and flat. Moreover, the square

$$\begin{tikzcd}
F_2 \arrow[r, "\widetilde{d}'_1"] \arrow[d, "\widetilde{d}'_0"'] & F_1 \arrow[d, "\widetilde{d}_0"] \\
F_1 \arrow[r, "\widetilde{d}_0"] & F
\end{tikzcd}$$

Figure 39: The Cartesian square used to descend W_1.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Exposé V, component-local p. 27 / combined-reader p. 289 / re-edited header p. 277.

[40] Editor's note: If $f : X \to Y$ is locally of finite type, the set of points $x \in X$ that are isolated in their fiber $f^{-1}(f(x))$ is open in X. In EGA III, 4.4.10, this is deduced from Zariski's Main Theorem when Y is locally Noetherian. For arbitrary Y, it follows from Chevalley's semicontinuity theorem (EGA IV$_3$, 13.1.3 and 13.1.4). Consequently, f is quasi-finite at x if and only if it is of finite type at x and x is isolated in $f^{-1}(f(x))$. This will be used below; see editor's note (42).

[41] Editor's note: The original has been expanded in what follows.

is Cartesian. It follows that

$$W_1 = \widetilde{d}_0^{-1}(U)$$

for an open subset U of F (SGA 1, VIII, 4.4). This open subset U is a quasi-section of the groupoid with base W induced by X_*. We may therefore take $U_z = W$.

It remains to state the following lemma, whose proof is classical.

Lemma 7.4.—*Consider a Cartesian square of schemes*

$$\begin{tikzcd} F_2 \arrow[r, "v"] \arrow[d, "d'"'] & F_1 \arrow[d, "d"] \\ X_1 \arrow[r, "u"] & X_0 \end{tikzcd}$$

Figure 40: The Cartesian square in Lemma 7.4.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Exposé V, component-local p. 27 / combined-reader p. 289 / re-edited header p. 277.

and let x be a point of F_2.

(i) If u is flat, d' is flat at x if and only if d is flat at $v(x)$.

(ii) If d is locally of finite type, d' is quasi-finite at x if and only if d is quasi-finite at $v(x)$.[42]

We have thus proved that X_0 has a covering by saturated open subsets W such that the groupoid W_* induced by X_* on W possesses a quasi-section.[43]

By Lemma 6.1 and the reductions stated after Theorem 7.1, this proves assertions (i) and (iii) of that theorem, as well as the facts that p is surjective and open and that p and $Y \to S$ are locally of finite presentation. Moreover, because $X_0 \to S$ is quasi-projective, and hence separated and of finite type, p is separated; the proof of Lemma 6.1(ii) shows that p and $Y \to S$ are of finite presentation.

To prove that p is proper, it remains to prove that it is universally closed. Since this assertion is local on Y, we may work over a saturated open subset W such that the groupoid W_* induced by X_* on W possesses a quasi-section U (because X_0 is covered by such open subsets). With the notation of 6(a), there is a commutative diagram

[42] Editor's note: The conditions are sufficient by base change (EGA II, 6.2.4(iii), and EGA IV_2, 2.1.4). Conversely, put $y = d'(x)$ and $z = u(y) = d(v(x))$, and suppose that d' is flat at x and that u (hence also v) is flat. Then $\mathcal{O}_{v(x)} \to \mathcal{O}_x$ is faithfully flat, as is $\mathcal{O}_z \to \mathcal{O}_y \to \mathcal{O}_x$. Consequently, $\mathcal{O}_z \to \mathcal{O}_{v(x)}$ is faithfully flat (EGA IV_2, 2.2.11(iv)). Finally, suppose that d is locally of finite type and that d' is quasi-finite at x. Then $v(x)$ is isolated in its fiber $d^{-1}(z)$, since x is isolated in its fiber $d'^{-1}(y) = d^{-1}(z) \otimes_{\kappa(z)} \kappa(y)$. By Chevalley's semicontinuity theorem there is therefore an open neighborhood of $v(x)$ every point of which is isolated in its fiber (EGA IV_3, 13.1.3 and 13.1.4), and hence d is quasi-finite at $v(x)$.

[43] Editor's note: The sequel has been modified to take advantage of the additions made to Lemma 6.1.

$$\begin{tikzcd} W \arrow[dr, "p"'] & V \arrow[l, "v"'] \arrow[r, "u"] \arrow[d, "r"] & U \arrow[dl, "q"] \\ & Z & \end{tikzcd}$$

Figure 41: The properness diagram over the open subset $Z \subseteq Y$.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Exposé V, component-local p. 28 / combined-reader p. 290 / re-edited header p. 278.

where Z is an open subset of Y, all the arrows are surjective, and q is integral. By hypothesis $d_0 : X_1 \to X_0$ is proper, so u, which is obtained from d_0 by base change, is proper as well. Consequently r is universally closed, and hence so is p, since v is surjective.

Finally, because p is surjective and universally closed and X_0 is quasi-projective, hence separated, the diagonal $\Delta_{Y/S}(Y)$ is closed in $Y \times_S Y$: it is the image under $p \times p$ of the diagonal $\Delta_{X_0/S}(X_0)$. Thus Y is separated over S. This completes the proof of Theorem 7.1(ii).

The assertions in Theorem 7.1(iv) are local on Y. Since the saturated open subsets U_z cover X_0, it is enough to verify them after replacing X and Y by U_z and $V_z = p(U_z)$, respectively. As observed at the beginning of the proof of Theorem 7.1, Lemmas 1.1, 1.2, and 6.1(i) show that $(V_z, p|_{U_z})$ is the cokernel, both in (**Sch**) and in the category of ringed spaces, of the groupoid induced by X_* on U_z. The hypothesis that $d = (d_0, d_1)$ is a monomorphism is preserved under the base change $U_z \to X_0$. The first two assertions of Theorem 7.1(iv) therefore follow from Lemma 6.1(iv).

Let us finally prove the consequences stated at the end of part (iv) (see [Ray67a], Theorem 1(iii)). By hypothesis, the groupoid X_* comes from an equivalence relation $R \to X_0 \times X_0$. We have proved that R is effective (Exposé IV, 3.3.2) and that $p : X_0 \to Y = X_0/R$ is faithfully flat and of finite presentation. Consequently, if (M) denotes the family of faithfully flat morphisms locally of finite presentation, then R is (M)-effective. By Exposé IV, 6.3.3, the pair (Y, p) therefore represents the quotient sheaf of X_0 by R for the fppf topology, and the assertions concerning base change follow from Exposé IV, 3.4.3.1.

8. Passage to the quotient by a flat groupoid that is not necessarily proper

Theorem 8.1.—[44] *Let S be a Noetherian scheme and*

[44] Editor's note: There is a largest open subset W of X_0 satisfying the conclusions of the theorem. Indeed, let W be an open subset as in the theorem and let $W^\sharp$ be a dense saturated open subset of W. Since p is open, $V^\sharp = p(W^\sharp)$ is open in V, and $W^\sharp = p^{-1}(V^\sharp)$, because $W^\sharp$ is saturated. By Lemma 1.1, $V^\sharp$ is a cokernel for the groupoid induced on $W^\sharp$. Thus two open subsets W and W' satisfying the conclusions of the theorem may be glued along their intersection $W^\sharp$, and conditions (i)–(iv), together with the property that p and $V \to S$ are locally of finite presentation, are preserved. Conclusion (ii') follows, as in the proof of Lemma 6.1(ii), from the hypothesis that X_0 is of finite type over the Noetherian scheme S.

Lemmas 1.1 and 1.2 also show that the union of all saturated open subsets $U \subseteq X_0$ such that the open subset $p(U) \subseteq Y$ is a scheme and $p|_U : U \to p(U)$ is a morphism of schemes is the largest saturated open subset $\Omega \subseteq X_0$ satisfying condition (i) of Theorem 8.1. The theorem shows that Ω contains a dense open subset W, but it is not immediate that Ω satisfies properties (ii)–(iv). For more precise results, and for the representability of the fppf quotient sheaf $\widetilde{X/R}$ (where R denotes the groupoid X_* with base $X = X_0$), see [Ray67a], [Ray67b], and Appendix I of [An73]. These works use weaker hypotheses: S may be arbitrary, X is locally of finite type over S, and R is an S-groupoid with base X such that d_0 (and hence d_1) is flat and

$$\begin{tikzcd} X_2 \arrow[r, shift left=2, "d'_2"] \arrow[r, "d'_1"] \arrow[r, shift right=2, "d'_0"'] & X_1 \arrow[r, shift left, "d_1"] \arrow[r, shift right, "d_0"'] & X_0 \end{tikzcd}$$

Figure 42: The $(\mathbf{Sch}/S)$-groupoid of Theorem 8.1.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Exposé V, component-local p. 29 / combined-reader p. 291 / re-edited header p. 279.

a $(\mathbf{Sch}/S)$-groupoid such that d_1 is flat and of finite type, X_0 is of finite type over S, and the morphism $X_1 \to X_0 \times_S X_0$ with components d_0 and d_1 is quasi-finite.

There exists a dense saturated open subset $W \subseteq X_0$ having the following properties:

(i) *if $W_2 \overset{w'_i}{\rightrightarrows} W_1 \overset{w_j}{\rightrightarrows} W$ is the groupoid induced by X_* on W, then (w_0, w_1) has a cokernel (V, p) in $(\mathbf{Sch}/S)$; moreover, (V, p) is a cokernel of (w_0, w_1) in the category of all ringed spaces;*

(ii) *p is surjective and open;*

(ii') *p and $V \to S$ are of finite presentation;*

(iii) *the morphism $W_1 \to W \times_V W$ with components w_0 and w_1 is surjective;*

(iv) *if (d_0, d_1) is an equivalence pair, then $W_1 \to W \times_V W$ is an isomorphism and p is faithfully flat.*

We shall prove that W may be chosen so that the $(\mathbf{Sch}/S)$-groupoid W_* induced by X_* possesses a quasi-section (see §7). Theorem 8.1 will then follow from Lemma 6.1.

Suppose provisionally that for every point $z \in X_0$ closed relative to S (see §7) there is a saturated open subset W_z that possesses a quasi-section and meets every irreducible component of X_0 passing through z. Then the complement $X_0 - W_z$ is saturated: its saturation $d_1(d_0^{-1}(X_0 - W_z))$ is open and does not meet W_z. If this complement is nonempty, choose in it a point z' closed relative to S, and associate to z' an open subset $W_{z'}$ as above. We may further suppose that $W_{z'} \subseteq X_0 - W_z$. Then W_z and $W_{z'}$ are disjoint, and the groupoid induced by X_* on $W_z \cup W_{z'}$ possesses a quasi-section. The process must stop, because X_0 has only finitely many irreducible components. It therefore remains to construct W_z.

For this purpose we may suppose that S is affine. Let $y \in X_1$ satisfy $d_1(y) = z$, let X be an affine open subset of X_0 containing $d_0(y)$, let Y be the inverse image of X under $d_0 : X_1 \to X_0$, and let $u : Y \to X$ and $v : Y \to X_0$ be the morphisms induced by d_0 and d_1. Since X is affine, hence quasi-projective, Lemma 7.2 provides a subscheme $F \subseteq X_0$ such that $d_0^{-1}(F) \cap d_1^{-1}(z)$ is nonempty, $d_0(d_0^{-1}(F) \cap d_1^{-1}(z))$ is finite, and the restriction of d_1 to $d_0^{-1}(F)$ is flat at the points of $v^{-1}(z)$. We can therefore resume the notation of Lemma 7.3, writing F_1 and F_2 for the inverse images of F in X_1 and X_2, and so on:

of finite presentation. In particular, if $\widetilde{X/R}$ is represented by an S-scheme Y, then Y is also the cokernel in the category of ringed spaces. The converse is false in general (see [Mum65], Chapter 0, §3, Example 0.4, cited in [Ray67a], Remark 1), but it is true when $d = (d_0, d_1)$ is an immersion. Under this hypothesis, $p : \Omega \to Z := \Omega/R_\Omega$ is faithfully flat and of finite presentation. If, in addition, S is locally Noetherian, then a codimension-one point x of X belongs to Ω if and only if the graph of the groupoid induced on $\operatorname{Spec}(\mathcal{O}_{X,x})$ is closed. See [Ray67a], Proposition 1; [Ray67b], Proposition 1 and Theorems 2, 1, and 4; and [An73], Theorems 5 and 6 on pages 66–67 and Proposition 3.3.1 on page 49. For an action of an algebraic group over an algebraically closed field k, see also [DR81].

$$\begin{tikzcd}
F_2 \arrow[r, shift left, "\widetilde{d}'_1"] \arrow[r, shift right, "\widetilde{d}'_0"'] \arrow[d, "\widetilde{d}_2"'] & F_1 \arrow[r, "\widetilde{d}_0"] \arrow[d, "\widetilde{d}_1"] & F \\
X_1 \arrow[r, shift left, "d_1"] \arrow[r, shift right, "d_0"'] & X_0 &
\end{tikzcd}$$

Figure 43: The induced diagram on F_2, F_1, F used in the proof of Theorem 8.1.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Exposé V, component-local p. 30 / combined-reader p. 292 / re-edited header p. 280.

As in 7.3, one proves that $\widetilde{d}_1$ is quasi-finite at the points of $\widetilde{d}_1^{-1}(z)$. It is therefore natural to consider the open subset $F'_1 \subseteq F_1$ consisting of the points at which $\widetilde{d}_1$ is both flat and quasi-finite. By Lemma 7.4, the inverse images of F'_1 under $\widetilde{d}'_1$ and $\widetilde{d}'_0$ both consist of the points of F_2 at which $\widetilde{d}'_2$ is flat and quasi-finite. Thus these inverse images coincide, and F'_1 is of the form $\widetilde{d}_0^{-1}(F')$ for an open subset $F' \subseteq F$ (SGA 1, VIII, 4.4). After replacing F by F', we may therefore suppose that $\widetilde{d}_1$ is quasi-finite and flat. Let W_z be the largest open subset of $\widetilde{d}_1(F_1)$ over which $\widetilde{d}_1$ is finite and flat.

The open subset W_z need not contain z, but it contains the generic points of the irreducible components of X_0 passing through z.[45] Since d_0 (and likewise d_1) is faithfully flat and of finite presentation, hence open, SGA 1, VIII, 5.7 shows that $d_0^{-1}(W_z)$ and $d_1^{-1}(W_z)$ both coincide with the largest open subset of $\widetilde{d}'_2(F_2)$ over which $\widetilde{d}'_2$ is finite and flat. As in 7.3, the inverse images of $\widetilde{d}_1^{-1}(W_z)$ under $\widetilde{d}'_0$ and $\widetilde{d}'_1$ therefore coincide. Hence $\widetilde{d}_1^{-1}(W_z) = \widetilde{d}_0^{-1}(U)$ for an open subset $U \subseteq F$, and U is a quasi-section for the groupoid induced by X_* on W_z.

9. Elimination of the Noetherian hypotheses in Theorem 7.1

a) Resume the notation and hypotheses of Lemma 6.1, and let $\pi : S' \to S$ be an arbitrary base change. If $f : X \to Y$ is a morphism of S-schemes, write $f' : X' \to Y'$ for the morphism of S'-schemes obtained from f by extension of the base along π. With this convention, $p' : X'_0 \to Y'$ is surjective, as is the morphism

$$X'_1 \longrightarrow X'_0 \times_{Y'} X'_0$$

with components d'_0 and d'_1. Thus the set underlying Y' is the quotient of the set underlying X'_0 by the equivalence relation on X'_0 defined by the S'-groupoid X'_*. Moreover, $q' : U' \to Y'$ is integral and surjective. The topology of Y' is consequently the quotient topology of that of U', and hence also of that of X'_0; see the proof in §6(c).

On the other hand, U' is plainly a quasi-section of the S'-groupoid X'_*, so Lemma 6.1 applies. In particular, X'_* has a cokernel (Y_1, p_1), and the topological space underlying Y_1 is obtained from that underlying X'_0 by identifying the points equivalent under the relation

[45] Editor's note: Indeed, let η be such a generic point. The hypotheses imply that $\mathcal{O}_{X_0,\eta}$ is an Artinian local ring and that $\mathcal{O}_{F_1,\eta}$ is a finitely generated $\mathcal{O}_{X_0,\eta}$-module. By SGA 1, VIII, 6.5, there is therefore an open neighborhood of η over which $\widetilde{d}_1$ is finite.

defined by X'_*. It follows that the canonical morphism $Y_1 \to Y'$ is a homeomorphism. In fact, it is a universal homeomorphism. Indeed, if S'' is an S'-scheme, let Y_2 be the cokernel of $(d_0 \times_S S'', d_1 \times_S S'')$. The preceding argument, applied to the base changes $S'' \to S'$ and $S'' \to S$, shows that

$$Y_2 \longrightarrow Y_1 \times_{S'} S'' \qquad \text{and} \qquad Y_2 \longrightarrow Y \times_S S'' \simeq Y' \times_{S'} S''$$

are homeomorphisms. Hence

$$Y_1 \times_{S'} S'' \longrightarrow Y' \times_{S'} S''$$

is a homeomorphism as well.

b) The same remarks plainly apply when the groupoid X_* *locally* possesses quasi-sections (see the proof of Theorem 7.1).[46] For example, we have the following theorem:

Theorem 9.0.—*Let S be an arbitrary scheme and*

$$X_2 \xrightarrow{d'_j} X_1 \xrightarrow{d_i} X_0$$

a $(\mathbf{Sch}/S)$-groupoid such that X_0 is of finite presentation and quasi-projective over S, d_1 is of finite presentation, proper, and flat, and the morphism

$$d_0 \boxtimes d_1 : X_1 \longrightarrow X_0 \times_S X_0$$

is quasi-finite. Then:

(1) *Every point x of X_0 has a saturated open neighborhood W such that the groupoid induced by X_* on W has a quasi-section.*

(2) *Let (Y, p) be the cokernel of (d_0, d_1) in the category of all ringed spaces. Then Y is a scheme, p is a morphism of schemes, and (Y, p) is a cokernel of (d_0, d_1) in $(\mathbf{Sch}/S)$.*

(3) *The morphism p is surjective, open, and universally closed.*

(4) *The morphism*
$$d : X_1 \longrightarrow X_0 \times_Y X_0$$
with components d_0 and d_1 is surjective.

(5) *If (d_0, d_1) is an equivalence pair, then:*

   (a) *the morphism*
   $$d : X_1 \longrightarrow X_0 \times_Y X_0$$
   is an isomorphism and p is faithfully flat;

   (b) *the morphisms p and $Y \to S$ are of finite presentation, and (Y, p) is a cokernel of (d_0, d_1) in the category of sheaves for the fppf topology.*

Proof. For (1), the question is local on S, so we may suppose that $S = \operatorname{Spec} B$ is affine. There then exist a ring A of finite type over $\mathbb{Z}$, a morphism

$$S \longrightarrow T = \operatorname{Spec} A,$$

[46] Editor's note: What follows has been expanded in order to bring out Theorem 9.0 below.

and a $(\mathbf{Sch}/T)$-groupoid Z_* such that

$$X_* \simeq Z_* \times_T S.$$

Indeed, apply EGA IV_3, 8.8.3 to $S_0 = \operatorname{Spec}\mathbb{Z}$ and $S_i = \operatorname{Spec} A_i$, where the A_i range over the finitely generated $\mathbb{Z}$-subalgebras of B. Moreover, by EGA IV_3, 8.10.5, we may suppose that Z_* satisfies the hypotheses of Theorem 7.1. Consequently, Z_* has quasi-sections locally.

By part (a), the same is therefore true of X_*, and assertions (2), (3), (4), and (5)(a) follow from Lemma 6.1, just as in the proof of Theorem 7.1.

c) Proof of (5)(b).—Let us show that $Y \to S$ is of finite presentation.[47] By hypothesis, $(d_0^{X_*}, d_1^{X_*})$ is an equivalence pair; that is,

$$d^{X_*} : X_1 \longrightarrow X_0 \times_S X_0$$

is a monomorphism. By EGA IV_3, 8.10.5, after enlarging A if necessary, we may suppose that

$$d^{Z_*} : Z_1 \longrightarrow Z_0 \times_T Z_0$$

is a monomorphism. Since $T = \operatorname{Spec} A$, with A Noetherian, Theorem 7.1 then shows that the groupoid Z_* has a cokernel (Q, q) in $(\mathbf{Sch}/T)$, that q and $Q \to T$ are of finite presentation, and, moreover, that

$$q : Z_0 \longrightarrow Q$$

is faithfully flat and d^{Z_*} induces an isomorphism

$$Z_1 \xrightarrow{\sim} Z_0 \times_Q Z_0.$$

Put $Q_S = Q \times_T S$. Since $X_i \simeq Z_i \times_T S$, we obtain an isomorphism

$$d^{Z_*} \times_T S : X_1 \xrightarrow{\sim} X_0 \times_{Q_S} X_0.$$

Denote its inverse by ϕ, and let π be the canonical morphism

$$X_0 \times_Y X_0 \longrightarrow X_0 \times_{Q_S} X_0.$$

Then $\phi \circ \pi$ is the inverse of

$$d_0 \boxtimes d_1 : X_1 \xrightarrow{\sim} X_0 \times_Y X_0.$$

It follows that the equivalence relation defined by X_*, that is, the monomorphism shown in Figure 44,

$$X_1 \xrightarrow[\sim]{d_0 \boxtimes d_1} X_0 \times_Y X_0 \hookrightarrow X_0 \times_S X_0$$

Figure 44: The equivalence-relation monomorphism defined by X_*.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Exposé V, component-local p. 33 / combined-reader p. 295 / re-edited header p. 283.

identifies with the equivalence relation $\mathscr{R}(q_S)$ defined by $q_S : X_0 \to Q_S$. Since q_S is faithfully flat and of finite presentation, hence a universal effective epimorphism, $\mathscr{R}(q_S)$ has quotient Q_S; see Exposé IV, 3.3.2. Consequently,

$$Y \simeq Q \times_T S,$$

[47] Editor's note: The original states that this follows from Proposition 9.1 below. We were unable to reconstruct that argument. The proof that follows was communicated to us by O. Gabber.

so $Y \to S$ and p are of finite presentation. Moreover, by Exposé IV, 6.3.3, (Y, p) is also a cokernel of (d_0, d_1) in the category of sheaves for the fppf topology.

Proposition 9.1.—*Consider morphisms of schemes*

$$X_0 \xrightarrow{p} Y \xrightarrow{q} S$$

such that qp is of finite type (respectively, of finite presentation) and p is faithfully flat and of finite presentation. Then q is of finite type (respectively, of finite presentation).[*]

Since p is surjective and qp is quasi-compact, q is quasi-compact. We may therefore suppose that S, Y, and X_0 are affine, with rings A, B, and C, respectively. We have

$$B = \varinjlim_i B_i,$$

where the B_i range over the finitely generated A-subalgebras of B. Since C is of finite presentation over B, there exist an index i_0, a B_{i_0}-algebra C_{i_0} of finite presentation, and an isomorphism

$$C \simeq C_{i_0} \otimes_{B_{i_0}} B.$$

For $i \geq i_0$, put

$$C_i = C_{i_0} \otimes_{B_{i_0}} B_i;$$

then

$$C \simeq C_i \otimes_{B_i} B.$$

$$\begin{tikzcd}
B \arrow[r] & C \\
B_i \arrow[u] \arrow[r] & C_i \arrow[u] \\
A \arrow[u] &
\end{tikzcd}$$

Figure 45: The compatible A-algebras used in the limit argument.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Exposé V, component-local p. 33 / combined-reader p. 295 / re-edited header p. 283.

Since C is faithfully flat over B, EGA IV_3, 11.2.6 and 8.10.5(vi) give an index $i_1 \geq i_0$ such that C_{i_1} is faithfully flat over B_{i_1}. Consequently, C_i is faithfully flat over B_i for every $i \geq i_1$. For $i \geq i_1$, the canonical map $C_i \to C$ is then injective, because it is obtained from $B_i \to B$ by faithfully flat base extension.

If C is of finite type over A, it follows that there is an index $j \geq i_1$ such that $C_j = C$, whence $B_j = B$, because C_j is faithfully flat over B_j. Thus B is of finite type over A.

Now suppose that C is of finite presentation over A. By what precedes, B is of finite type over A, and hence has the form

$$B = \overline{B}/I,$$

[*] See EGA IV_4, 17.7.5 for a more general result.

where $\overline{B}$ is a polynomial algebra over A in finitely many indeterminates and I is an ideal of $\overline{B}$. The ideal I is the union of its finitely generated subideals I_α; consequently,

$$B = \varinjlim_{\alpha} B_\alpha, \qquad B_\alpha = \overline{B}/I_\alpha.$$

Proceeding as above, there exist an index α_0, a B_{α_0}-algebra C_{α_0} of finite presentation, and an isomorphism

$$C \simeq C_{\alpha_0} \otimes_{B_{\alpha_0}} B.$$

For $\alpha \geq \alpha_0$, put

$$C_\alpha = C_{\alpha_0} \otimes_{B_{\alpha_0}} B_\alpha,$$

so that

$$C \simeq C_\alpha \otimes_{B_\alpha} B \qquad (\alpha \geq \alpha_0).$$

Again by EGA IV_3, 11.2.6 and 8.10.5(vi), C_α is faithfully flat over B_α for all sufficiently large α. In that case, the kernels of the maps $C_\alpha \to C$ and $C_\alpha \to C_\beta$, respectively, identify with

$$C_\alpha \otimes_{B_\alpha} (I/I_\alpha) \quad \text{and} \quad C_\alpha \otimes_{B_\alpha} (I_\beta/I_\alpha) \qquad (\beta \geq \alpha).$$

Since C_α and C are of finite presentation over A and $C_\alpha \to C$ is surjective,

$$C_\alpha \otimes_{B_\alpha} (I/I_\alpha)$$

is a finitely generated ideal,[48] and it is the union of the ideals

$$C_\alpha \otimes_{B_\alpha} (I_\beta/I_\alpha).$$

Thus, for all sufficiently large β,

$$C_\alpha \otimes_{B_\alpha} (I_\beta/I_\alpha) = C_\alpha \otimes_{B_\alpha} (I/I_\alpha).$$

Since C_α is faithfully flat over B_α, this also gives $I_\beta = I$. Therefore B is of finite presentation over A.

10. Supplement: quotients by a group scheme

Sections 10.2–10.4 below, written following suggestions of M. Raynaud, apply the preceding theorems to the case of an action of a group scheme. For the reader's convenience, Section 10.1 first reproduces statements 2.1–2.3 of Exposé XVI.

10.1. Representability theorems for quotients.—We first recall the following result.

Theorem 10.1.1.—*Let S be a scheme, let X and Y be two S-schemes, and let $f : X \to Y$ be an S-morphism. Suppose that one of the following two conditions holds:*

α) *the morphism f is locally of finite presentation;*

β) *the scheme S is locally Noetherian and X is locally of finite type over S.*

Then the following conditions are equivalent:

[48] Editor's note: see EGA IV_1, 1.4.4.

(i) *there exist an S-scheme X' and a factorization*

$$f : X \xrightarrow{f'} X' \xrightarrow{f''} Y,$$

where f' is a faithfully flat S-morphism locally of finite presentation and f'' is a monomorphism;

(ii) *the first projection*

$$p_1 : X \times_Y X \longrightarrow X$$

is flat.

Moreover, if these conditions hold, then (X', f') is a quotient of X by the equivalence relation defined by f for the fppf topology. Thus the factorization $f = f'' \circ f'$ in (i) is unique up to isomorphism.

The case in which Y is locally Noetherian and X is of finite type over Y is treated in [Mur65], Corollary 2 to Theorem 2. We shall see that the general case reduces to that one.

We first make a few remarks.

a) The implication (i)⇒(ii) is immediate. Indeed, the first projection

$$p'_1 : X \times_{X'} X \longrightarrow X$$

factors through $X \times_Y X$:

$$p'_1 : X \times_{X'} X \xrightarrow{u} X \times_Y X \xrightarrow{p_1} X.$$

The morphism u is an isomorphism because f'' is a monomorphism, and p'_1 is flat because f' is flat; hence p_1 is flat.

b) The assertions of Theorem 10.1.1 are local on Y, and hence local on S. They are also local on X, as follows readily from the fact that a flat morphism locally of finite presentation is open (EGA IV_3, 11.3.1).

c) Under hypothesis α of Theorem 10.1.1, the preceding remarks reduce us to the case where X and Y are affine and f is of finite presentation. Replacing S by Y, if necessary, we may suppose that X and Y are of finite presentation over S. EGA IV_3, 11.2.6 then reduces us to the case where S is Noetherian.

d) Under hypothesis β of Theorem 10.1.1, we may suppose that S, X, and Y are affine, that S is Noetherian, and that X is of finite type over S. Write Y as a filtered inverse limit of affine schemes Y_i of finite type over S. The schemes

$$X \times_{Y_i} X$$

form a filtered decreasing family of closed subschemes of $X \times_S X$, whose inverse limit is $X \times_Y X$. Since $X \times_S X$ is Noetherian, for all sufficiently large i we have

$$X \times_{Y_i} X = X \times_Y X.$$

It follows that

$$f_i : X \xrightarrow{f} Y \longrightarrow Y_i$$

satisfies condition (ii) of Theorem 10.1.1 whenever f does. The equivalence relation on X defined by f is the same as the one defined by f_i. Thus it suffices to prove (ii)⇒(i) for f_i, reducing us to the case where Y is of finite type over S.

Application to group schemes. Let S be a scheme, let G be an S-group scheme locally of finite presentation over S, and suppose that G acts on the left on an S-scheme X. If $X \to S$ has a section ξ, recall that the stabilizer $\mathrm{Stab}_G(\xi)$ is representable by a subgroup scheme of G; see Exposé I, 2.3.3.

Theorem 10.1.2.—*Let S be a scheme and let G be an S-group scheme locally of finite presentation over S, acting on an S-scheme X. Suppose that $X \to S$ has a section ξ whose stabilizer H in G is flat over S. If one of the following hypotheses holds:*

a) X is locally of finite type over S;

b) S is locally Noetherian,

then the fppf quotient sheaf G/H is representable by an S-scheme, locally of finite presentation over S, and the S-morphism

$$f : G \longrightarrow X, \qquad g \longmapsto g \cdot \xi$$

factors as

$$\begin{tikzcd} G \arrow[d, "p"'] \arrow[dr, "f"] & \\ G/H \arrow[r, hook, "i"] & X \end{tikzcd}$$

Figure 46: The factorization of the orbit morphism in Theorem 10.1.2.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Exposé V, component-local p. 36 / combined-reader p. 298 / re-edited header p. 286.

where p is the canonical projection, which is faithfully flat and locally of finite presentation, and i is a monomorphism.

Proof.— The morphism f makes G into an X-scheme. By definition of the stabilizer of ξ, the morphism

$$G \times_S H \longrightarrow G \times_X G, \qquad (g, h) \longmapsto (g, gh)$$

is an isomorphism. Since H is flat over S, the scheme $G \times_S H$ is flat over G; hence the first projection

$$p_1 : G \times_X G \longrightarrow G$$

is flat.[49] Moreover, if X is locally of finite type over S, then f is locally of finite presentation (EGA IV_1, 1.4.3(v)); otherwise, S is assumed locally Noetherian. It therefore suffices to apply Theorem 10.1.1 to f. It remains only to see that G/H is locally of finite presentation over S, and this follows immediately from Proposition 9.1.

Corollary 10.1.3.— *Let S be a scheme and $u : G \to H$ a morphism of S-group schemes. Suppose that G is locally of finite presentation over S, and that either H is locally of finite type over S or S is locally Noetherian.*

[49] The source PDF prints $G \times_S G$ in this projection. The preceding displayed isomorphism and the application of Theorem 10.1.1 require the fiber product $G \times_X G$, which is the reading adopted here.

If $K = \ker(u)$ is flat over S, then the quotient group G/K is representable by an S-group scheme locally of finite presentation over S, and u factors as

$$\begin{tikzcd} G \arrow[rr, "u"] \arrow[dr, "p"'] & & H \\ & G/K \arrow[ur, "i"'] & \end{tikzcd}$$

Figure 47: The factorization of u through G/K.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Exposé V, component-local p. 36 / combined-reader p. 298 / re-edited header p. 286.

where p is the canonical projection and i is a monomorphism.

Proof.— Apply Theorem 10.1.2 with $X = H$ and with the unit section of H as ξ.

10.2. Stabilizer of the diagonal

Let S be a Noetherian scheme, X an S-scheme of finite type, and G a flat S-group scheme of finite type acting on the left on X. Thus we have an S-action $d_0 : G \times_S X \to X$. Let $d_1 : G \times_S X \to X$ be projection onto the second factor. As in §2(a), we have the groupoid

$$\begin{tikzcd} G \times_S G \times_S X \arrow[r, shift left=2, "\mathrm{pr}_{2,3}"] \arrow[r, "\mu \times X" description] \arrow[r, shift right=2, "G \times d_0"'] & G \times_S X \arrow[r, shift left, "d_1"] \arrow[r, shift right, "d_0"'] & X \end{tikzcd}$$

Figure 48: The action groupoid attached to (X, G).

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Exposé V, component-local p. 36 / combined-reader p. 298 / re-edited header p. 286.

whose cokernel, when it exists, is denoted $G\backslash X$.

Definition 10.2.1.— *We denote by $F \subset G \times_S X$ the* stabilizer of the diagonal section*; that is, F is the X-scheme defined by the Cartesian square*

$$\begin{tikzcd} F \arrow[r] \arrow[d] & X \arrow[d, "\Delta"] \\ G \times_S X \arrow[r, "(d_0,d_1)"] & X \times_S X \end{tikzcd}$$

Figure 49: The stabilizer F of the diagonal section.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Exposé V, component-local p. 37 / combined-reader p. 299 / re-edited header p. 287.

Then F is an X-subgroup scheme of $G \times_S X$. Since $G \times_S X$ is of finite type over the Noetherian scheme S, it is Noetherian; consequently F is of finite type both over S and over

X (EGA I, 6.3.5 and 6.3.6). If $X \to S$ is separated, then F is a closed X-subgroup scheme of $G \times_S X$.

Recall that G is said to *act freely* on X when

$$(d_0, d_1) : G \times_S X \longrightarrow X \times_S X$$

is a monomorphism (Exposé III, 3.2.1). Equivalently, F is the trivial group scheme over X.

10.3. The case in which F is quasi-finite over X

Since F is of finite type over X, it is quasi-finite over X if and only if the stabilizers of the geometric points of X are finite.

Theorem 10.3.1.—[50] *Under the hypotheses of 10.2, suppose that F is quasi-finite over X. Then there is a dense, G-saturated open subset $U \subset X$ having the following properties:*

(i) *in ($\mathbf{Sch}/S$), the cokernel $V = G\backslash U$ exists; moreover, V is a quotient in the category of ringed spaces;*

(ii) *$p : U \to V$ is surjective, open, and of finite presentation;*

(iii) *V is of finite presentation over S;*

(iv) *the morphism*

$$G \times_S U \longrightarrow U \times_V U, \qquad (g, x) \longmapsto (gx, x),$$

is surjective;

(v) *if, in addition, G acts freely on X, then $U \to V$ is a left G-torsor, locally trivial for the fppf topology. In particular, $U \to V$ is faithfully flat.*[51]

Proof.— By hypothesis, the morphism

$$G \times_S X \longrightarrow X \times_S X, \qquad (g, x) \longmapsto (gx, x),$$

is quasi-finite. Theorem 8.1 therefore applies to the groupoid defined by (X, G). Hence there is a dense saturated open subset $U \subset X$ for which the quotient $G\backslash U$ exists, and it has properties (i), (ii), and (iii); the same theorem also gives (iv).

To prove (v), recall that G acts freely on X if and only if (d_0, d_1) is an equivalence pair (Exposé III, 3.2.1).[52] In that case Theorem 8.1(iv) shows that

$$G \times_S U \longrightarrow U \times_V U$$

is an isomorphism and that p is faithfully flat and of finite presentation. Thus U is a G-torsor over V, locally trivial for the fppf topology.

[50] Editor's note: here too there is a largest open subset $U \subset X$ satisfying the conclusions of the theorem; see editor's note 44.

[51] Editor's note: if one assumes in addition that G is a reductive S-group scheme and that its free action on X is linearizable, then $G\backslash X$ is known to be representable and $X \to G\backslash X$ is a left G-torsor. This follows from results of Raynaud and Seshadri and appears in [CTS79], Proposition 6.11.

[52] At this point the source PDF says *to prove (iv)*. The cited Theorem 8.1(iv) and the conclusion about a torsor establish part (v) in the present numbering; part (iv) follows from Theorem 8.1(iii).

10.4. The case in which F is flat over X

Write
$$d = (d_0, d_1) : G \times_S X \longrightarrow X \times_S X, \qquad d(g, x) = (gx, x).$$
Recall that the sheaf-theoretic graph $\widetilde{\Gamma}$ of the equivalence relation associated with (X, G) is the fppf S-subsheaf of $X \times_S X$ given by the image of (d_0, d_1). It is the fppf sheaf associated with the graph functor
$$T \longmapsto \Gamma(T) = \big\{(x_0, x_1) \in X(T) \times X(T) \mid x_0 \in G(T)x_1\big\}.$$
Put $G_X = G \times_S X$. For every S-scheme T there is a surjective map
$$G_X(T) \longrightarrow \Gamma(T), \qquad (g, x) \longmapsto (gx, x),$$
which induces a bijection
$$\phi(T) : G_X(T)/F(T) \xrightarrow{\sim} \Gamma(T).$$
Indeed, if $(g, x), (g', x') \in G_X(T)$ have the same image, then $x' = x$ and $g^{-1}g'x = x$, so that $(g^{-1}g', x) \in F(T)$, and (g, x) and (g', x) have the same image in $G_X(T)/F(T)$.[53]

By definition (see Exposé IV, 4.4.1(ii), or the proof of 5.2.1), the quotient sheaf G_X/F is the fppf sheaf associated with the presheaf
$$T \longmapsto G_X(T)/F(T) \simeq \Gamma(T).$$
We therefore obtain an isomorphism of sheaves
$$\phi : G_X/F \xrightarrow{\sim} \widetilde{\Gamma}.$$

Theorem 10.4.1.—[54] *Under the hypotheses of 10.2, the following assertions hold:*

a) the sheaf $\widetilde{\Gamma}$ is representable if and only if F is flat over X;

b) suppose that F is flat over X. Then the morphisms induced by d_1 and d_0

$$G_X/F \underset{\overline{d}_0}{\overset{\overline{d}_1}{\rightrightarrows}} X$$

Figure 50: The two maps induced by d_1 and d_0.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Exposé V, component-local p. 38 / combined-reader p. 300 / re-edited header p. 288.

are faithfully flat and of finite presentation.

[53] The source PDF prints $(g^{-1}g'x, x) \in F(T)$. Since $F(T) \subseteq G(T) \times X(T)$, the first coordinate must be the group element $g^{-1}g'$; the English adopts this type-correct reading. In the same sentence the source writes $F_X(T)$ once in the denominator; the displayed bijection immediately above and the presheaf immediately below both use the defined stabilizer $F(T)$, which is retained consistently here.

[54] Editor's note: this is part (2) of Theorem 3 of [Ray67b]. That note also sketches another proof of Theorem 10.1.1.

Proof of a).— Suppose that the fppf sheaf G_X/F is represented by an X-scheme Y. Then, by Exposé IV, 6.3.3, the morphism $p : G_X \to Y$ is faithfully flat and locally of finite presentation, and the second square in the following diagram is Cartesian:

$$\begin{tikzcd}
F \arrow[r] \arrow[d] & F \times_X G_X \arrow[r] \arrow[d] & G_X \arrow[d, "p"] \\
X \arrow[r, "e_X"] & G_X \arrow[r] & Y
\end{tikzcd}$$

Figure 51: The Cartesian square used in the proof of Theorem 10.4.1(a).

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Exposé V, component-local p. 39 / combined-reader p. 301 / re-edited header p. 289.

The first square is obtained by base change along the unit section $e_X : X \to G_X$. Since p is faithfully flat and locally of finite presentation, the same is true of $F \to X$.

Conversely, suppose F is flat over X, and put $X^2 = X \times_S X$. The morphism $d : G_X \to X^2$ gives the fiber square

$$\begin{tikzcd}
G_X \times_{X^2} G_X \arrow[r] \arrow[d] & G_X \arrow[d] \\
G_X \arrow[r] & X^2
\end{tikzcd}$$

Figure 52: The fiber square associated with $d : G_X \to X^2$.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Exposé V, component-local p. 39 / combined-reader p. 301 / re-edited header p. 289.

The first projection

$$\mathrm{pr}_1 : G_X \times_{X^2} G_X \longrightarrow G_X$$

is then an $F \times_X G_X$-torsor over G_X, and is therefore flat and of finite type, because F is.[55] By Theorem 10.1.1, d factors uniquely as

$$G_X \xrightarrow{\psi} Y \xrightarrow{\tau} X \times_S X$$

Figure 53: The unique factorization of d.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Exposé V, component-local p. 39 / combined-reader p. 301 / re-edited header p. 289.

where ψ is faithfully flat and of finite type, and τ is a monomorphism of schemes.

[55] The source PDF instead prints the morphism $G_X \times_{X^2} G_X \to X^2$ and calls it an $F \times_X X^2$-torsor over X^2. That morphism is not generally surjective. The first projection to G_X is the stabilizer torsor and is the flatness input required by Theorem 10.1.1; the English uses this type-correct reading.

It follows that the morphism of sheaves $\psi : G_X \to Y$ is F-invariant, and hence induces a morphism

$$\bar{\psi} : G_X/F \longrightarrow Y.$$

Moreover, because ψ is faithfully flat and of finite type, the monomorphism of sheaves τ factors through the sheaf image of d, namely $\widetilde{\Gamma}$. Thus the isomorphism $G_X/F \simeq \widetilde{\Gamma}$ factors through the monomorphism $Y \to \widetilde{\Gamma}$. We conclude that Y represents G_X/F.

Proof of b).— Suppose F is flat over X. By part (a) and its proof, G_X/F is representable, and

$$p : G_X \longrightarrow G_X/F$$

is faithfully flat and of finite presentation. On the other hand, the morphisms $d_i : G_X \to X$, for $i = 0, 1$, are faithfully flat and of finite presentation by hypothesis. Since $d_i = \bar{d}_i \circ p$, it follows from EGA IV_2, 2.2.13(iii), and EGA IV_3, 11.3.16, that each $\bar{d}_i$ is faithfully flat and of finite presentation.

Theorem 10.4.2.—[56] *Under the hypotheses of 10.2, suppose that F is flat over X. Then there is a dense saturated open subset $U \subset X$ such that the fppf quotient $V = G\backslash U$ is an S-scheme of finite type and $U \to V$ is faithfully flat and of finite presentation.*

Proof.— Theorem 10.4.1 shows that $G_X/F \simeq \widetilde{\Gamma}$ is representable. The fppf sheaf $G\backslash X$ is therefore the quotient sheaf of

$$G_X/F \overset{\bar{d}_1}{\underset{\bar{d}_0}{\rightrightarrows}} X$$

Figure 54: The equivalence pair whose quotient is $G\backslash X$.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Exposé V, component-local p. 40 / combined-reader p. 302 / re-edited header p. 290.

By what precedes,

$$\bar{d}_i : G_X/F \longrightarrow X$$

is faithfully flat and of finite presentation for $i = 0, 1$, and the morphism

$$G_X/F \xrightarrow{\ \sim\ } \widetilde{\Gamma} \hookrightarrow X \times_S X$$

Figure 55: The monomorphism from the quotient G_X/F to $X \times_S X$.

Native LaTeX reconstruction from the corrected Polo–Gille diagram; source locator: Exposé V, component-local p. 40 / combined-reader p. 302 / re-edited header p. 290.

is a monomorphism. In other words, $(\bar{d}_0, \bar{d}_1)$ is an equivalence pair. Consequently, Theorem 8.1 applies. There is therefore a dense saturated open subset $U \subset X$ such that the fppf quotient $V = G\backslash U$ is an S-scheme of finite type and $U \to V$ is faithfully flat and of finite presentation.

[56] Editor's note: here too there is a largest open subset $U \subset X$ satisfying the conclusions of the theorem. Moreover, a codimension-one point $x \in X$ belongs to U if and only if

$$(G_X/F) \times_X \mathrm{Spec}(\mathcal{O}_{X,x}) \longrightarrow \mathrm{Spec}(\mathcal{O}_{X,x}) \times_S \mathrm{Spec}(\mathcal{O}_{X,x})$$

is a closed immersion; see editor's note 44.

Taking the generic-flatness theorem into account (EGA IV_2, 6.9.3), we obtain the following result.

Corollary 10.4.3.— *Under the hypotheses of 10.2, suppose that X is reduced. Then there is a dense saturated open subset $U \subset X$ such that the fppf quotient $G\backslash U$ is an S-scheme of finite type and $U \to G\backslash U$ is faithfully flat and of finite presentation.*

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[57] Editor's note: additional references cited in this exposé.